APPLICATION EXAMPLE

AC500 V3 WEBVISUALIZATION

DEMONSTRATION EXAMPLE

Contents

1 Disclaimer

A. For customers domiciled outside Germany /

Für Kunden mit Sitz außerhalb Deutschlands

„Warranty, Liability:

The user shall be solely responsible for the use of this products described within this file. ABB shall be under no warranty whatsoever. ABB's liability in connection with application of the products or examples provided or the files included within this products, irrespective of the legal ground, shall be excluded. The exclusion of liability shall not apply in the case of intention or gross negligence. The present declaration shall be governed by and construed in accordance with the laws of Switzerland under exclusion of its conflict of laws rules and of the Vienna Convention on the International Sale of Goods (CISG)."

„Gewährleistung und Haftung:

Der Nutzer ist allein für die Verwendung des in diesem Dokument beschriebenen Produkte und beschriebenen Anwendungsbeispiele verantwortlich.

ABB unterliegt keiner Gewährleistung. Die Haftung von ABB im Zusammenhang mit diesem Anwendungsbeispiel oder den in dieser Datei enthaltenen Dateien - gleich aus welchem Rechtsgrund - ist ausgeschlossen. Dieser Ausschluss gilt nicht im Falle von Vorsatz oder grober Fahrlässigkeit. Diese Erklärung unterliegt Schweizer Recht unter Ausschluss der Verweisungsnormen und des UN-Kaufrechts (CISG)."

B. Nur für Kunden mit Sitz in Deutschland

„Gewährleistung und Haftung:

Die in diesem Dokument beschriebenen Anwendungsbeispiele oder enthaltenen Dateien beschreiben eine mögliche Anwendung der AC500 bzw. zeigen eine mögliche Einsatzart. Sie stellen nur Beispiele für Programmierungen dar, sind aber keine fertigen Lösungen. Eine Gewähr kann nicht übernommen werden.

Der Nutzer ist für die ordnungsgemäße, insbesondere vollständige und fehlerfreie Programmierung der Steuerungen selbst verantwortlich. Im Falle der teilweisen oder ganzen Übernahme der Programmierbeispiele können gegen ABB keine Ansprüche geltend gemacht werden.

Die Haftung von ABB, gleich aus welchem Rechtsgrund, im Zusammenhang mit den Anwendungsbeispielen oder den in dieser Datei enthaltenen Beschreibung wird ausgeschlossen. Der Haftungsausschluss gilt jedoch nicht in Fällen des Vorsatzes, der groben Fahrlässigkeit, bei Ansprüchen nach dem Produkthaftungsgesetz, im Falle der Verletzung des Lebens, des Körpers oder der Gesundheit oder bei schuldhafter Verletzung einer wesentlichen Vertragspflicht. Im Falle der Verletzung einer wesentlichen Vertragspflicht ist die Haftung jedoch auf den vertragstypischen, vorhersehbaren Schaden begrenzt, soweit nicht zugleich ein anderer der in Satz 2 dieses Unterabsatzes erwähnten Fälle gegeben ist. Eine Änderung der Beweislast zum Nachteil des Nutzers ist hiermit nicht verbunden.

Es gilt materielles deutsches Recht unter Ausschluss des UN-Kaufrechts."

2 Introduction

2.1 Scope of the document

This document describes the steps for several visualization features. These are for example a user input, animations, alarm management, the displaying of arrays, working with files, configuring multilanguage, working with recipes, traces and trends and enabling a user management, …

2.2 Compatibility

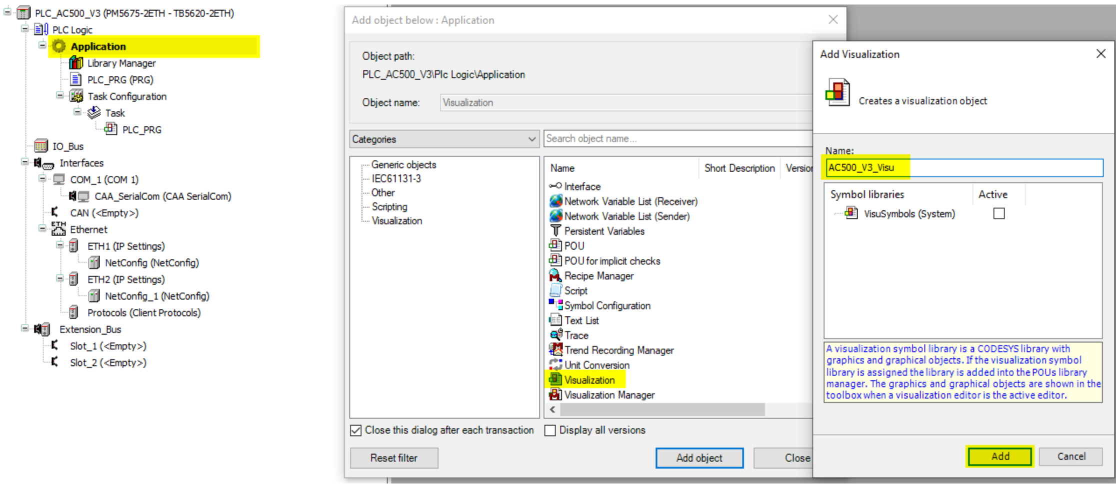
The application example explained in this document has been used with the below engineering system versions. They should also work with other versions, nevertheless some small adaptations may be necessary, for future versions.

- Automation Builder 2.5.0 or newer
- Simulation mode (limited features) or AC500 V3

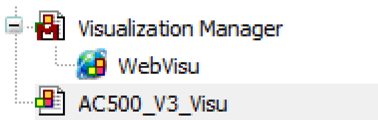
3 Configuration of a Web visualization

The following chapters describe the configuration of the AC500 V3 PLC to enable the web visualization and show a page in the web browser.

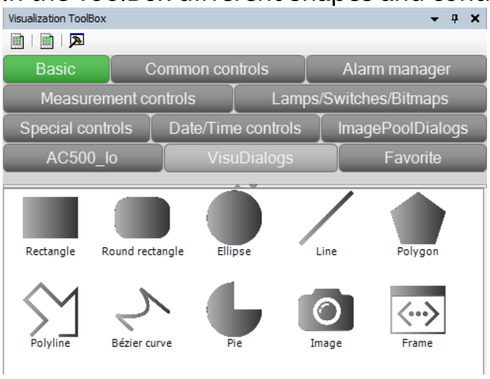
3.1 Visualization



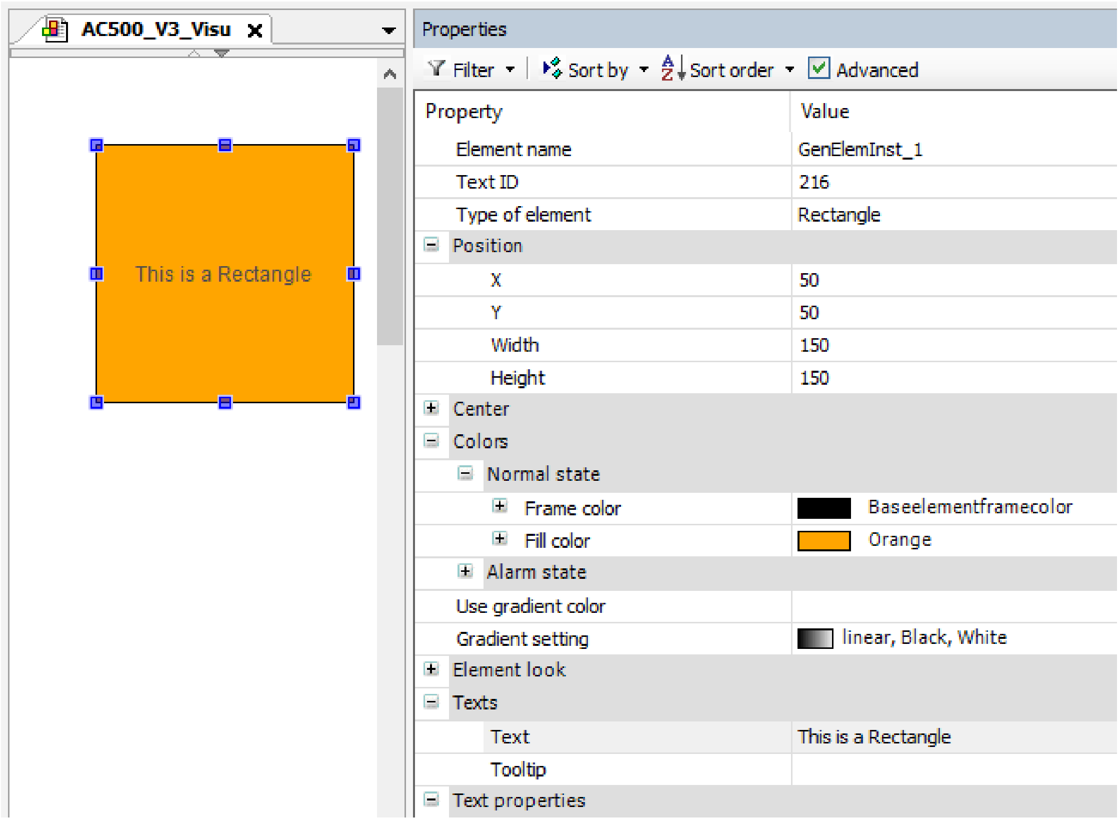
By right clicking on Application an object can be added. In this case a Visualization named AC500_V3_Visu is added below the application.



By adding the first visualization the Visualization Manager and the WebVisu is added as well to the device. The settings for these objects are explained in the next chapters.
In the ToolBox different shapes and controls can be added to the visualization by drag & drop.

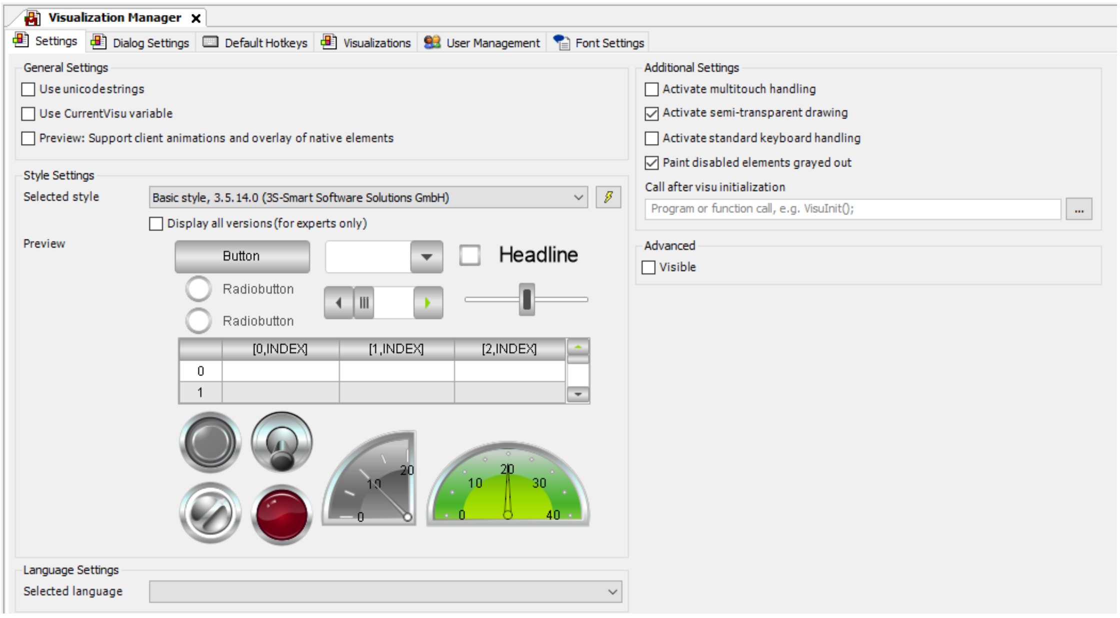


Once an element of the Visualization Toolbox is added to the visualization (i.e. Rectacngle) the properties of this element can be changed.
As shown in the picture below the Rectangle is positioned on X and Y coordinates 50. Width and height were set to 150. The Frame color is the Baseelementframecolor and the Fill Color is Orange. In the section Texts ‘This is a rectangle’ is written which is then shown as text inside the square.



3.2 Visualization Manager

The Visualization Manager is added together with the first visualization to the project.



In the General Settings the string type can be selected. By default, ASCII strings are used. In case it is needed to display also Unicode characters for example for Chinese language, the first check box can be selected. If the programmer needs to know which visualization is active in the moment the CurrentVisu variable can be used. This CurrentVisu can only be used inside Automation Builder and not in the webvisu.

All elements like Textboxes, Buttons, Sliders, Lamps and so on have a default layout. This is defined in the style. Usually, the Basic style is used. But if another style fits better to the Application you can change the style here. All elements in all applications change their style according to this setting. Further settings for the Style editor are explained in chapter 4.

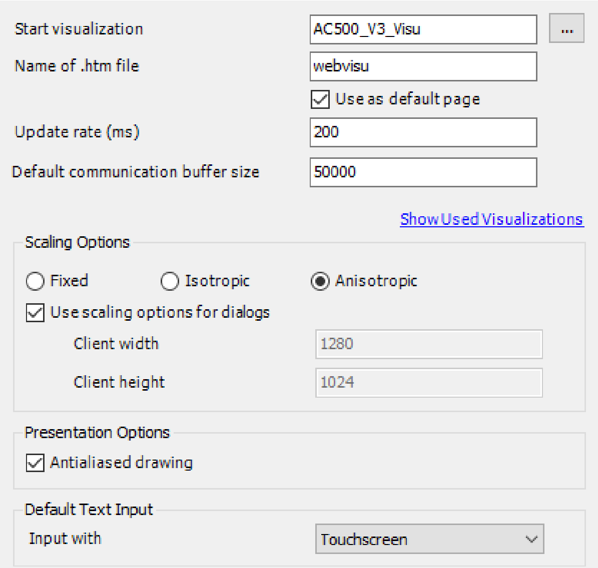
If a display with multitouch is used the activate multitouch handling setting can be checked. This allows to scroll, swipe and zoom in several widgets. In case of having multiple clients where not all clients have a multitouch display this setting can be enabled. The clients which don’t have the multitouch need to call the visualization with following parameter:

http(s)://<IpAddress>/webvisu.htm**?CFG_TouchHandlingActive=False**

With AB2.5.0 it is possible to detect for Linux devices, if the connected device has the multitouch functionality. Further settings which can be done in the visualization manager are available in the online help.

3.3 WebVisu

Below the Visualization Manager the WebVisu is located.

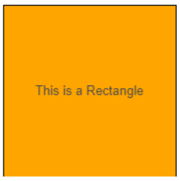


In the first row the user can select which visualization shall be used as start visualization. Below it is possible to specify the name of the default page the update rate as well as the buffer size.

In the scaling are three possibilities

- Anisotropic means the page is zoomed as big as possible and fills the complete screen. The aspect ratio might change.

- Isotropic means the aspect ratio of the pages remains but it is zoomed to show the complete page as big as possible.



- Fixed means the width and height which is specified is used. No scaling is used. Scrollbars are appearing in case not the full visualization is shown

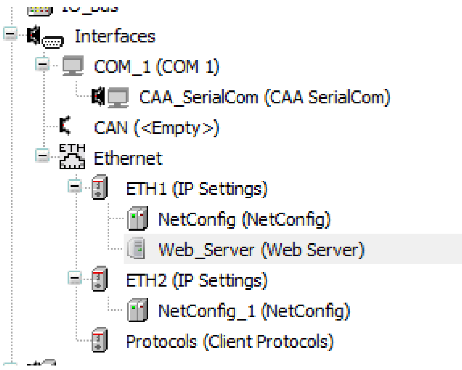
It is also possible to have more than one WebVisu object below the visualization manager. This way it is possible to have different start visualizations, scaling options or standard inputs for different devices. Only one WebVisu object can have the “Use as default page” box checked.

Different clients can call the different WebVisu start pages by different http links like

- http(s)://<IpAddress>/webvisu.htm default
- http(s)://<IpAddress>/webvisu_panel.htm for a special panel
- http(s)://<IpAddress>/webvisu_QR.htm for mobile devices via QR code

3.4 Webserver

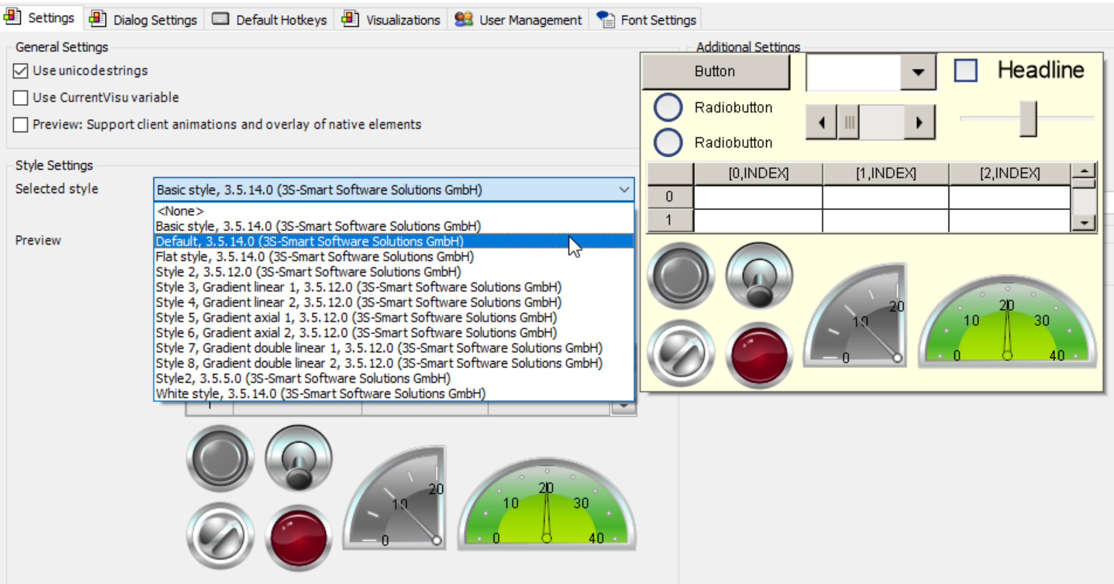
To display the pages also on a web browser or web panel a webserver has to be added to the ethernet configuration. As required to ETH1, ETH2 or both. By default, port 80 is used as HTTP Port and 443 as HTTPS Port.



If secure communication via HTTPS shall be used a server certificate is required. Furthermore, it is recommended to forward requests on port 80 to the HTTPS port 443. More details about security can be found in the white paper - Cyber Security in the AC500 PLC family. Please also check the Cyber Security FAQs.

4 Style settings

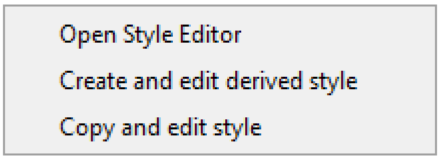
In the Visualization Manager the style settings can be changed. These settings are applied to all visualization elements. Several predefined styles are available which can be selected.



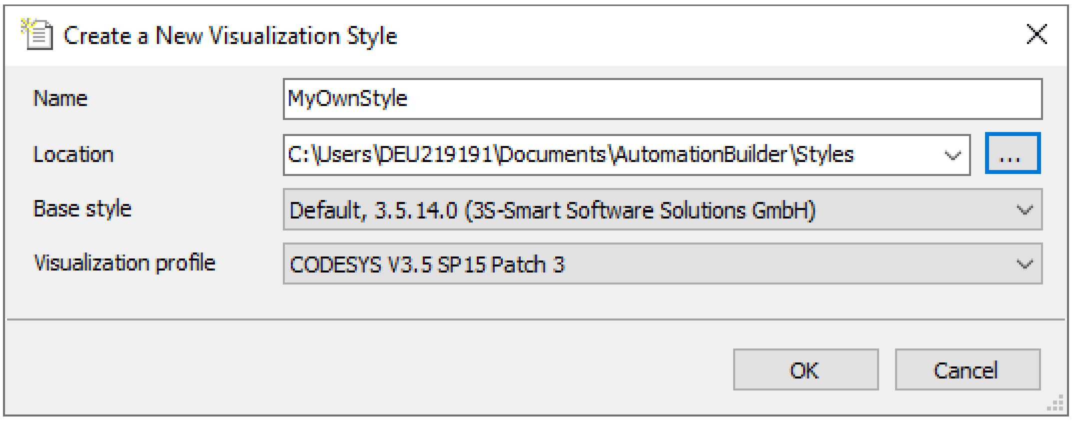
It is also possible to create a new style. Therefore, click on the  next to the drop-down menu. How to create your own style is described in this chapter.

4.1 Create a new style from a template

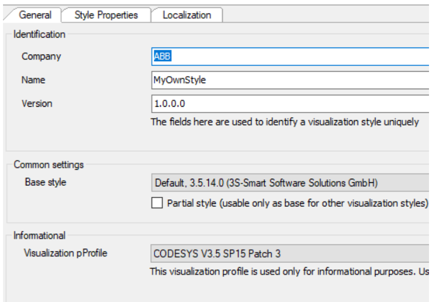
To create a new style, click  and select “Create and edit derived style”.



It would be also possible to open the style editor and create a new style from scratch or to copy an existing style and do the modifications which are needed. But the recommended option is to create a new style which is derived from another style.



Like shown above the style name and its location is specified. Important is also the base style which is derived. In the General settings which are automatically opened further settings can be done.

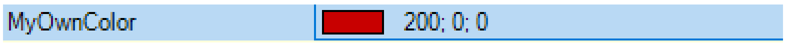


In the tab Style Properties the settings can be done to modify the style settings according to the needs like a company branding.

4.2 Colors

In the colors section several predefined colors are listed with its RGB values.

To add a new color, you can select “Colors” click . The color “New” with the default value Black is added. The name and the RGB values can be changed.



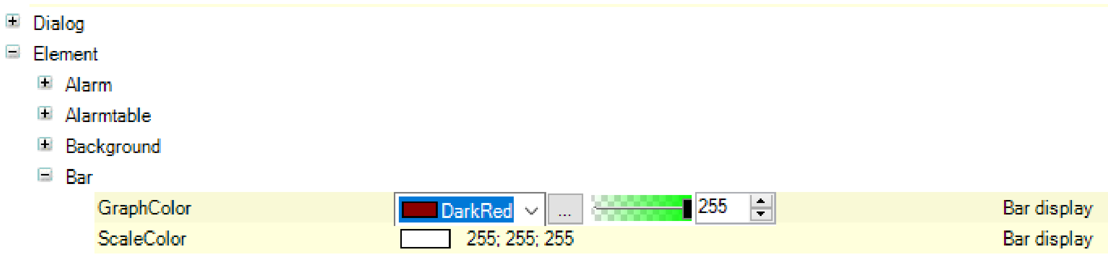
Another possibility is not to add a new color but to change the values from any existing.

The next step is to change the colors from existing elements.

In the list are BasicElement, Dialog, Element, Font and Progressbar. Inside of Element even more objects are listed.

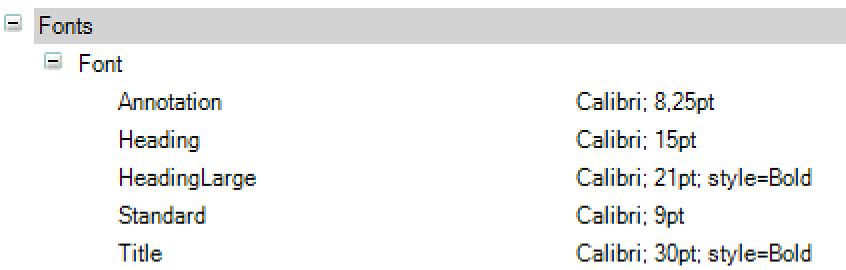
As example the “Bar” in “Elements” is chosen. From the default style the Graph Color is green.

With the drop-down menu it can be changed to any other color here Dark Red or the RGB values can be set manually. Similar to this the color of all elements and texts can be changed according to the requirements.



4.3 Fonts

In Fonts the different font types for the different elements can be changed.



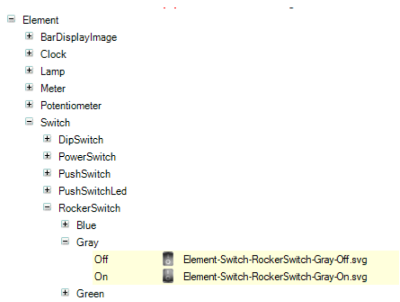
In this example the fonts are changed from Arial to Calibri. Also the size or the style could be changed.

It is also possible to add another Font here similar to the colors.

To change the default color of the texts please change the color of Fonts in the Color section.

4.4 Images

Many elements in the visualization like for example a switch have images in the background. Depending on the position of a switch it can be selected between two images. One for Off and another for On. If needed it is possible to select own pictures for all these elements.

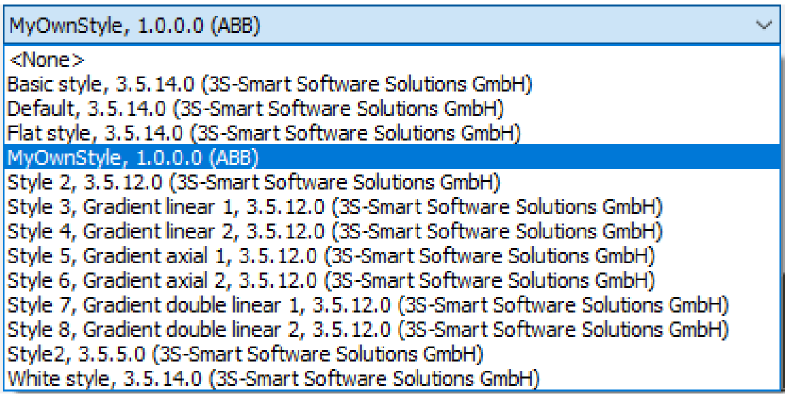


4.5 Installing and usage of a style

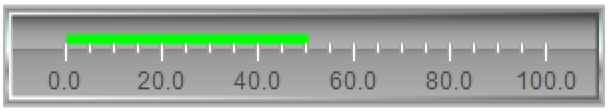
After adapting all the changes, the style can be saved. 

Afterwards the new style can be installed into the repository. 

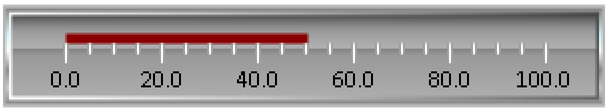
The new style is now available in the Drop-Down list in the Visualization Manager.



After selecting “MyOwnStyle” the look of Bar element which was modified is now changed as visible below.

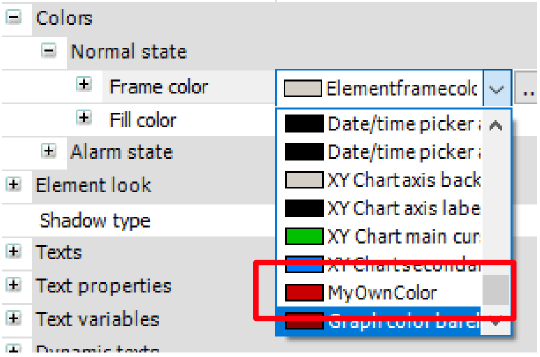


Default Style



My own Style

The color “MyOwnColor” which was additional added is now available in the drop down menu.



4.6 Further information

The style is locally installed to your PC. In case you want to share the style with colleagues they need to install the style also into their style repository. This can be done via Tools > Visualization Style Repository. The *.visustyle.xml file can be selected and installed.

In case you want to modify the style even further you can either open the style editor from Automation Builder and open your own style inside or start the style editor directly from the installation path:
C:\Program Files (x86)\ABB\AutomationBuilder\Common\VisualStylesEditor.exe

It is not possible to open the style editor by double clicking the *.visustyle.xml file.

Further information about the style editor are available in the online help:

PLC Automation with V3 CPUs > Programming with IEC 61131-3 editor > CODESYS Visualization > Applying visualization styles > Editing visualization styles in the visualization style editor

5 Display Values and User Input

The most common requirement is to display variables in the visualization and change these by a user input. Here three examples for boolean variables, integer and strings are explained.

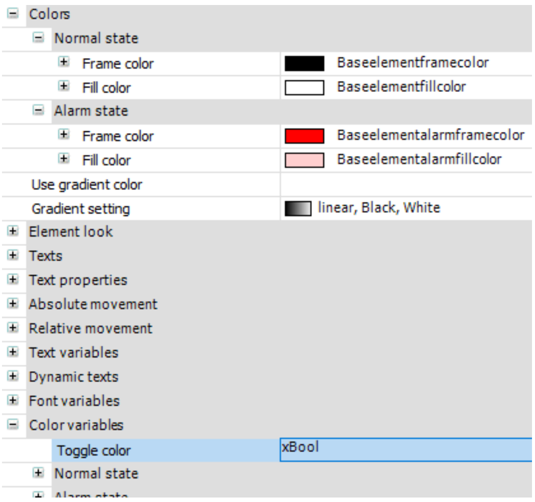
Everything described in this chapter can be found in the Visualization Sandbox which is not part of the web visualization

5.1 Boolean Variables

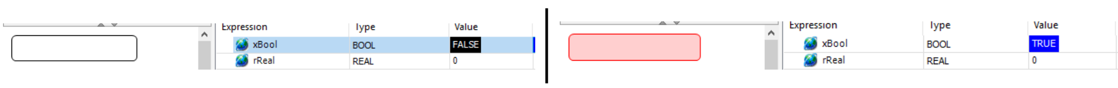
Different elements can be used to display the state of a boolean variables.

Select a Basic element from the Visualization Toolbox and drag it into your page. In this case a rounded rectangle is chosen.

The section Color of this element provides the possibility to change the Normal and the Alarm state color.



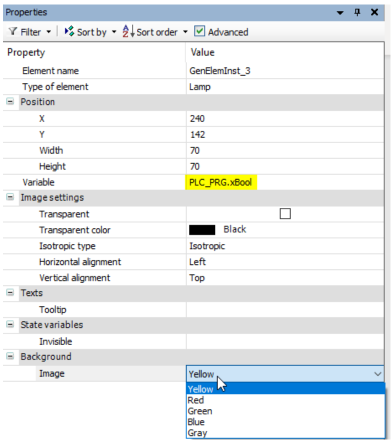
As toggle color a boolean variable can be used. If the variable is True, the Alarm color is shown instead of the normal color.



All Basic Elements can change their layout depending on a variable.

In the Lamps/ Switches/ Bitmaps a Lamp can also be used to show the Boolean state.

As Variable the variable in the program is chosen. In Background Image the color of the Lamp can be selected. The result by toggling the variable xBool in the program is shown below.


Properties
Filter
Sort by
Sort order
Advanced
Property
Value
Element name
GenElemInst_3
Type of element
Lamp
Position
X
240
Y
142
Width
70
Height
70
Variable
PLC_PRG.xBool
Image settings
Transparent
Transparent color
Black
Isotropic type
Isotropic
Horizontal alignment
Left
Vertical alignment
Top
Texts
Tooltip
State variables
Invisible
Background
Image
Yellow
Yellow
Red
Green
Blue
Gray

If there is the need to change a variable from the program the user has different possibilities. Next to the lamp different switches are located.

Again, the Boolean variable from the program is used. In the state variables this dip switch can also be hide, or its inputs deactivated depending on other states.

Property	Value
Element name	GenElemInst_4
Type of element	Dip switch
⊟ Position	
X	402
Y	173
Width	70
Height	70
Variable	PLC_PRG.xBool
⊟ Image settings	
Transparent	☐
Transparent color	Black
Isotropic type	Isotropic
Horizontal alignment	Left
Vertical alignment	Top
Element behavior	Image toggler
⊟ Texts	
Tooltip	
⊟ State variables	
Invisible	
Deactivate inputs	
⊟ Background	
Image	Gray

Similar to the switches a check box which is part of the common controls can be used.

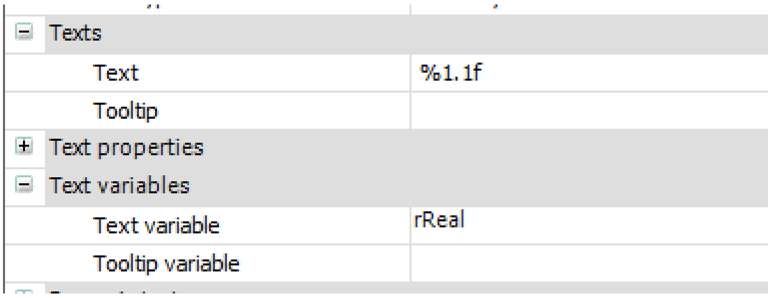
Another common control object for binary values is a button. In the section Input configuration, it can be specified which variable shall be taped and or toggled by pressing the button.

⊟ Input configuration	
OnDialogClosed	Configure...
OnMouseClick	Configure...
OnMouseDown	Configure...
OnMouseEnter	Configure...
OnMouseLeave	Configure...
OnMouseMove	Configure...
OnMouseUp	Configure...
⊟ Tap	
Variable	xBool
Tap FALSE	
Tap on enter if captured	
⊞ Toggle	
⊞ Hotkey	

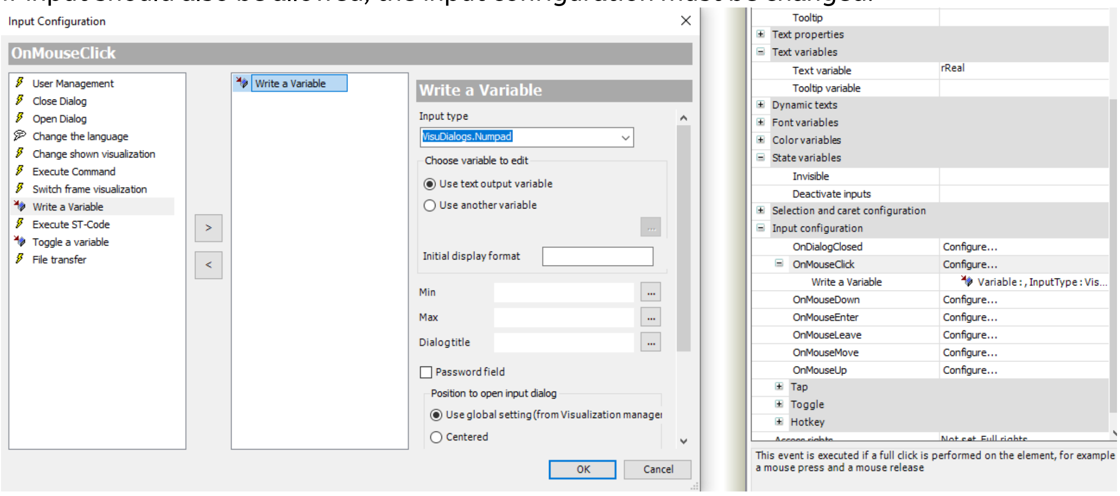
5.2 Integer or Real Variables

The widgets in the Measurement controls section can be used to display an integer or real variable. If also a user input should be possible a Slider can be chosen. It is furthermore possible to display the number itself and manipulate it with a numpad.

To display a number a textbox is added to the visualization. Depending on displaying an integer variable or a real variable %i or %f is used as Text. In Text variable the reference to the program variable is entered.



If input should also be allowed, the input configuration must be changed.



On a Mouse Click a Variable should be written. Use input type to select how the new number is entered. Here the numpad as visu dialog is chosen. Below it can be selected if the number which is displayed shall be changed or another variable. Furthermore, the display format of the variable as well as Min and Max values and the Dialog title can be defined.

5.3 Strings and Messages

Displaying and handling with strings is done with a textbox too, similar to the numbers. The shown Text is %s and the string variable is referenced as Text Variable. In the input configuration the Keypad is used instead of the Numpad.

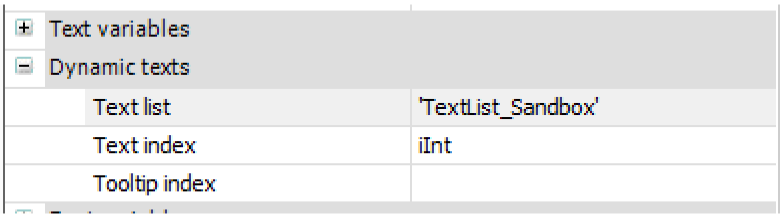
It is also possible to display messages. In the project an integer variable is used and depending on the value different messages shall be shown. For example, the state of a machine state is displayed. The possible values as well as the meaning is shown below.

State	0	1	2	3
Message	Init	Ready	Running	Error

In this case a TextList can be added to the application. In this text list the messages above can be entered with the right ID.

ID	Default
3	Error
2	Running
1	Ready
0	Init

In the visualization a text box is used which has a dynamic text. The text list is used and the corresponding integer variable from the program.



These dynamic texts have the advantage that existing variables from the program can be used without changing an additional string variable in different places in the program. Furthermore, all messages are located in one list and can be changed easily. In addition, text lists can also be translated which allows have Multilanguage see chapter 12.

5.4 Input Limitations

In many applications where a user input is possible there are limits on the entered values. The visualization shall prevent, that the user enters not allowed values. In this chapter, three possibilities are described how to archive this. Also Benefits and disadvantages are highlighted.

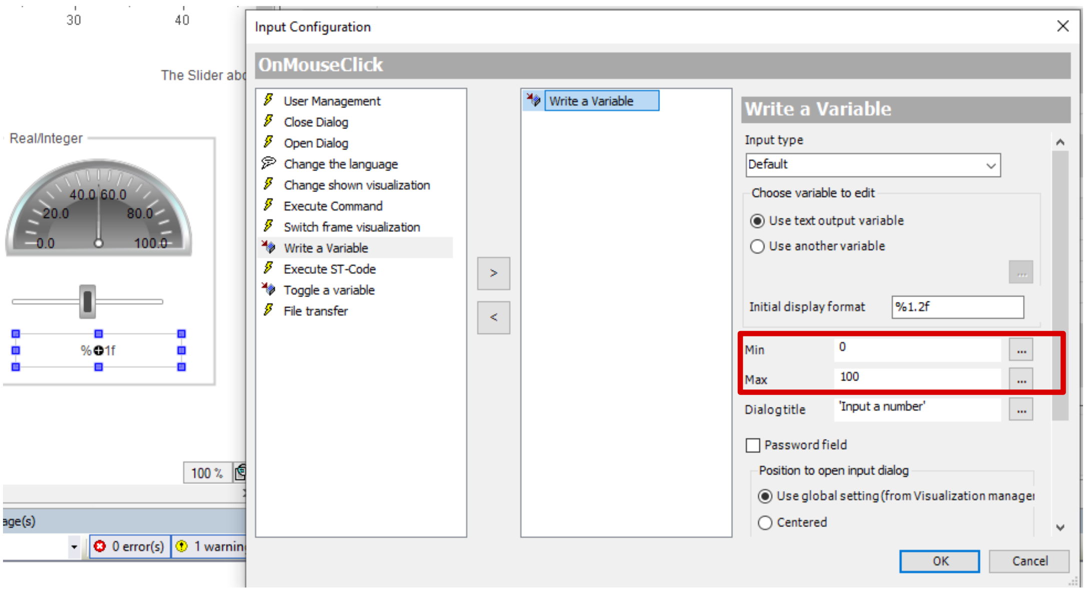
The simplest input limitation can be done in the element configuration. For example, a slider can have a min and max number for the scale. This way values out of the range cannot be used as an input.

5.4.1 Input Limitations in the action write a variable

Assuming there is a textbox linked to a real variable.

In the textbox properties an On Mouse Click Action is added. Here Write a Variable is chosen.

In the settings the Min and Max Variable can be specified. It is also possible to have dynamic variables for Min and Max inserted here.



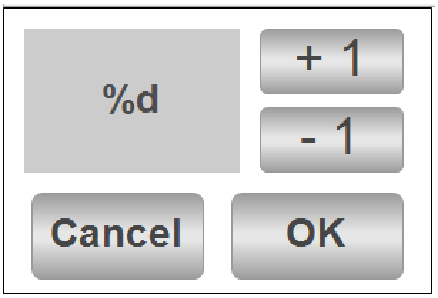
If now the user tries to enter a value which is out of range the input is not allowed. It is also not possible to input a character or a string as the variable itself has the type *REAL.*

This solution can be implemented very easy and fast. But only Min and Max values can be configured. More complex checks are not possible with this solution.

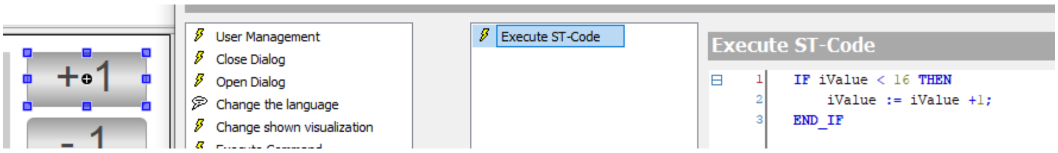
5.4.2 Limitations by using a dialog

In this chapter some settings for the button configuration in a dialog page are described. The handling with dialogs is described in the chapter 6.3.

The use case should be an integer input which has to be in the range between 0 and 16 where only even numbers are possible. In the dialog page two buttons +1 and -1 are used to change the number. Both buttons execute ST code when clicked.



The +1 button has a check inside the ST code. The increment is only executed when the value is below 16.



The -1 button doesn’t have this check inside the ST code. Here always an increment is executed. But the button itself can be disabled depending on a condition. If the value is below or equal zero the button is deactivated, cannot be clicked anymore and is greyed out.

⊟ State variables	
Invisible	
Deactivate inputs	iValue<=0
⊞ Button state variable	
⊞ Bitmap ID variable	
⊟ Input configuration	
OnDialogClosed	Configure...
⊟ OnMouseClick	Configure...
Execute ST-Code	iValue := iValue -1;
OnMouseDown	Configure...

Similar to this configuration also the OK button is configured. This buttons inputs can only be used if the variable iValue is even and between 0 and 16. Otherwise the button is deactivated.

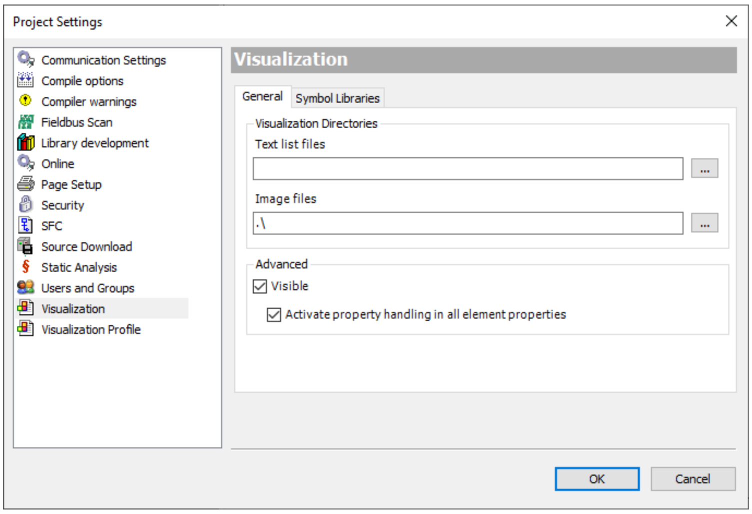
⊟ State variables	
Invisible	
Deactivate inputs	((iValue MOD 2)=1) OR iValue<=0 OR iValue >16
⊞ Button state variable	

In addition to the input limitations by writing a variable, which was shown in the chapter before, the user has more possibilities in this case. But another visualization as dialog is needed in the project which needs to be configured. If more complex checks need to be done, like checking for special numbers or string operations which need more than one line of code, properties have to be used.

5.4.3 Limitations by using properties

This solution is the most complex solution.

The visualization check by properties has to be enabled in the project settings. Please open Project → Project Settings and select the tab Visualization.



Here at first the advanced settings need to be checked as visible. Then activate property handling in all element properties must be activated.

	Note: By using this setting, the load of your system will increase. Furthermore, this setting might cause issues with other visualization elements.

In this project this setting is not activated. The two examples which will follow are inside the visualization Input_Limitations. This visualization slide is not included into the build. To use this visualization the checkbox in the Project settings has to be set and the Visualization has to be included into the Build by changing the parameter in the visualization properties.

As an example, there are two variables in a Global Variables list.

The integer variable must only have 2^n as value: 1, 2, 4, 8, 16, 32, … 16384. The access to this GVL can be controlled by a Property. A Property with Get and Set Method is added to the GVL. The same can also be done for a Program.

In the Get Method of the property the local variable is assigned to the property for reading the value.

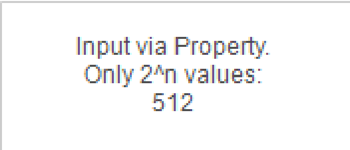
```
IntInputCheck := iIntVar;
```

The main logic is then done in the set Method of the property. Here a loop is used to compare the entered value with all 2^n values in the integer range. If the values are identical the entered value fits the requirements and is written to the local variable. If not, the loop ends without any findings. Then the local variable is not updated.

```
FOR iLoop := 0 TO 14 BY 1 DO
    IF IntInputCheck = iCompare THEN
        iIntVar := IntInputCheck; //This is a multiple of 2 assign variable
        RETURN; //Stop here comparing, found
    END_IF
    iCompare := iCompare * 2; //check next

END_FOR
// Loop finished without any results --> leave existing value as it is.
// IF there shall be an error handling this can be done here like assigning a default variable
```

In the visu it is possible to enter 512.



But when trying to enter for example 3 the shown value and also the internal value remains 512.

Similar to the handling with the integer variable also a property check for a String variable can be added.

Here for example the path for a file shall be entered. To avoid that the file is written into a wrong memory location all entered strings shall start with *sdcard/*. Only the extensions .txt and .csv are allowed. Spaces are also forbidden. Furthermore, all entered characters shall become lowercase.

Again, a Property is used to limit the input. The get method just outputs the local variable. The Set Method contains the logic.

```
VAR
    sToWorkWith : STRING;
    iTmp : INT;
    iTmp2 : DINT;
END_VAR
VAR CONSTANT
    sStart : STRING := 'sdcard/';
    sExtension1 : STRING := '.txt';
    sExtension2 : STRING := '.csv';
    sSpace : STRING := ' ';
END_VAR
```

```
sToWorkWith := PathInputCheck;
StrToLowerA(ADR(sToWorkWith)); //all characters to lower case
//Check if the string starts with sdcard
IF StrFindA(ADR(sToWorkWith), ADR(sStart), 1) <> 1 THEN //Not found at first position in the string --> stop
    RETURN;
END_IF
IF StrFindA(ADR(sToWorkWith), ADR(sExtension1),1) <> StrLenA(ADR(sToWorkWith))-3 THEN
    //Not Ending with .txt
    IF StrFindA(ADR(sToWorkWith), ADR(sExtension2),1) <> StrLenA(ADR(sToWorkWith))-3 THEN
        //Also not ending with .csv
        RETURN; //Abort
    END_IF
END_IF
IF StrFindA(ADR(sToWorkWith), ADR(sSpace), 1) <> 0 THEN
    RETURN; // 0 = Not found Abort if insed independend where it is
    //Could also be in a loop and working with a replace by _ or similar
END_IF

sStringVar := sToWorkWith;
```

At first the Property Value is copied into another local string variable to avoid a pointer to pointer access. With the String function ToLower all characters in the sToWorkWith string are set to lower case.

In the next condition it is checked if the string *sdcard/* can be found at the first position of the entered string so if this string starts with sdcard. In the next condition it is check if the extension is either .txt or .csv. If these strings cannot be found at the right position Return is called to abort the Property Method without any return.

In the last step it is also checked if there is a space inside the array. If yes return is called.

If the Set Method of the property was not aborted at any condition it ends with assigning the local variable in the GVL which is then also updated in the visu.

If one of these conditions is not fulfilled the text in the visualization will not change.

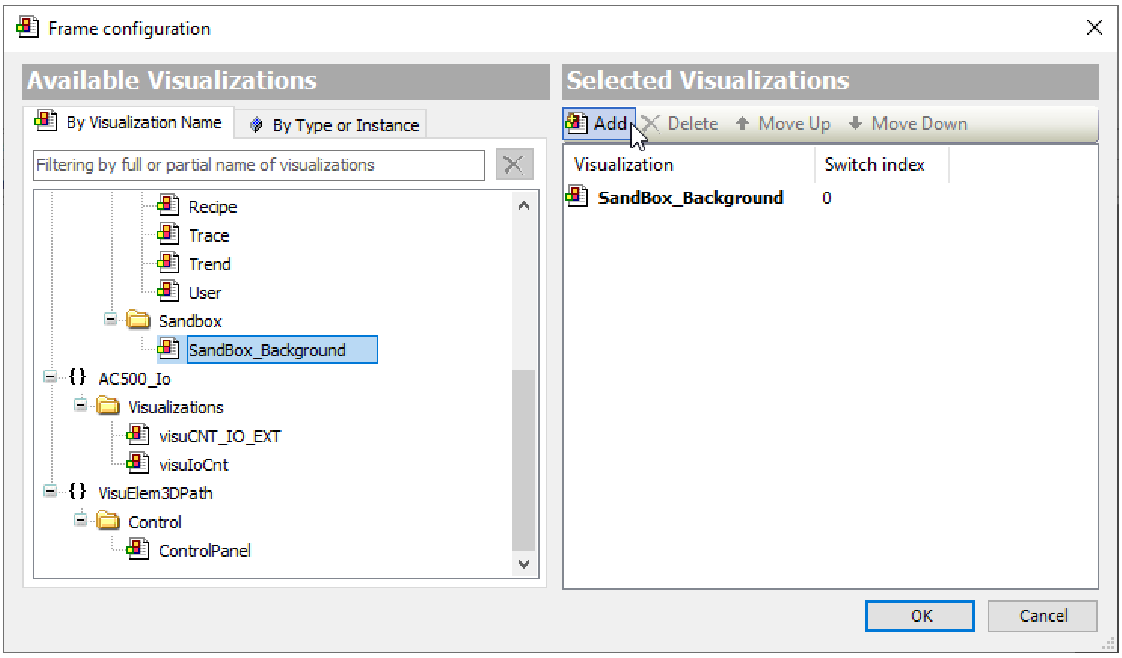
6 Page Management

6.1 Background Pages

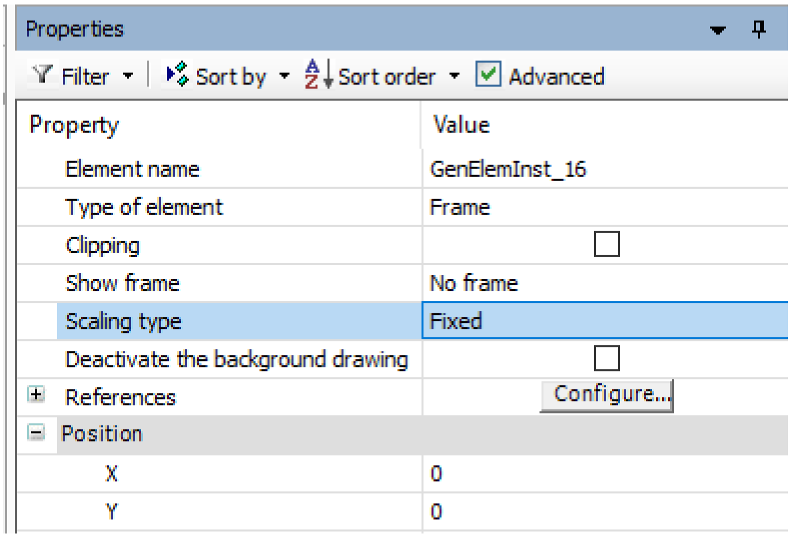
Background pages as known from AC500 V2 or Panel Builder Projects are not possible in the AC500 V3 Web visualization. It is only possible to have a background color or image.

A new possibility is to have visualization pages inside other visualizations. This way also a background visualization can be configured. In this example the visualization page SandBox_Background, which is already existing, shall be used as background page for the visualization Sandbox.

In the Sandbox Visualization, add a frame from the Basic topic. Select the Background Page in the pop-up window and click Add.



Please make sure that both visualizations have the same size. Then select the scaling type Fixed. Set the X and Y Position to 0.



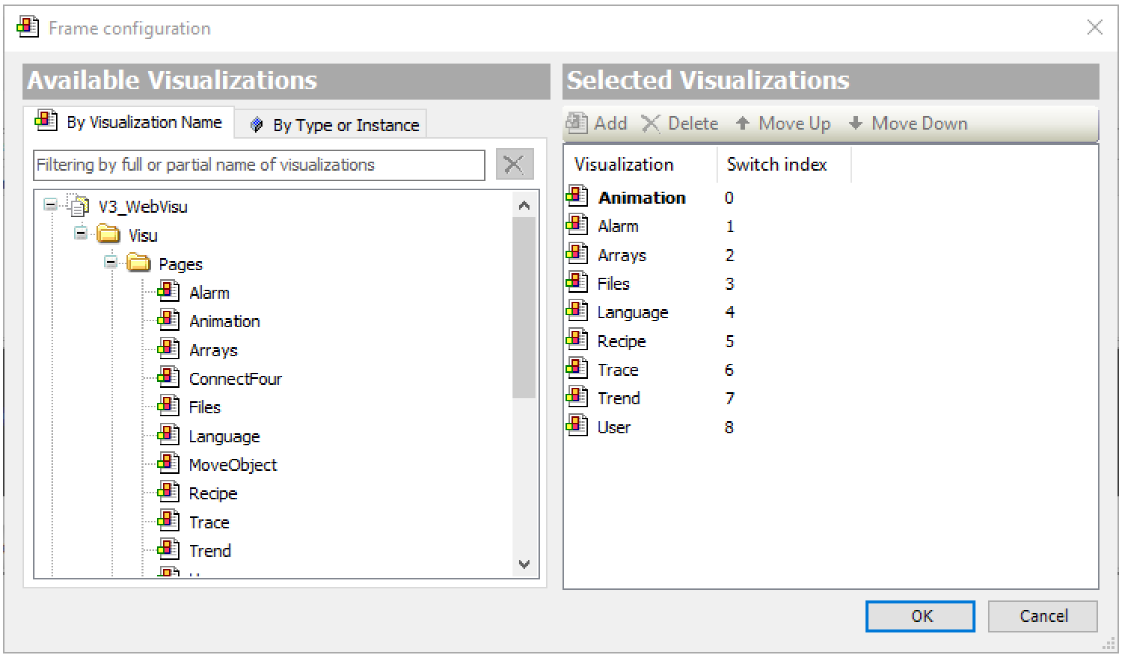
If there are already objects on this page, Right click on the Background frame and select Order, Send to Back. Now the other Visualization is in the Background. But as it is only a frame it is possible to move this frame or change its size.

This background page can then be copied to all other visualizations, where the background is needed.

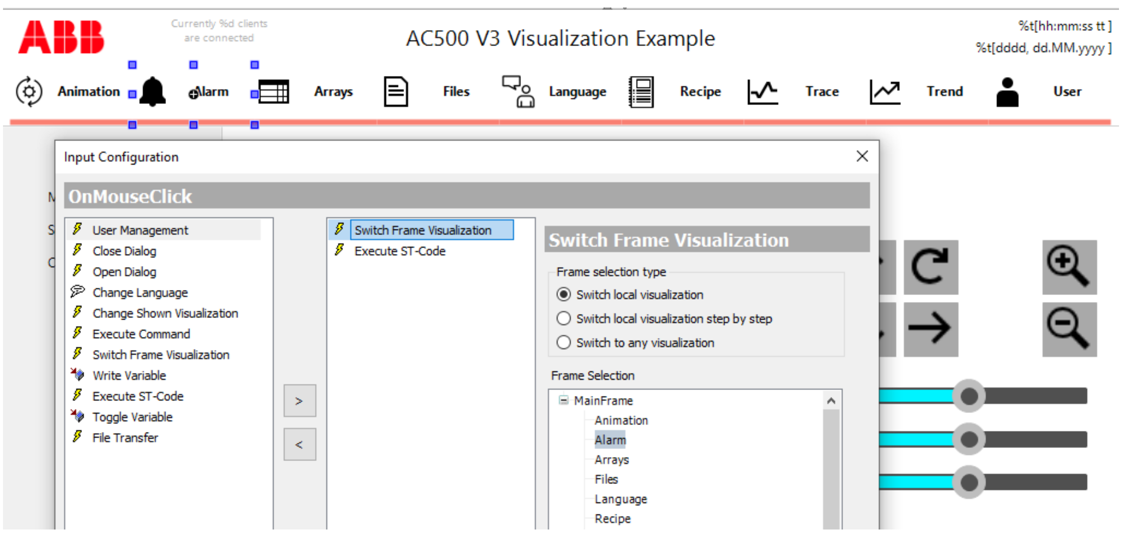
Instead of using a background as described here it is possible to have multiple pages inside a frame. This is described in the next chapter.

6.2 Frames & Tab Control

With Frames and the Tab control you can have Pages inside Pages. Similar to the background Page more than one Visualization can be added to the Frame. Therefore, refer to the Visualization AC500_V3_Visu. Here a Frame with multiple Visualizations inside is added.



To change the shown visualization different Buttons are used. The buttons have a on Mouse Click action which is changing the frame visualization.



It is also possible to switch the visualization step by step. Here the user can configure buttons as First, Previous, Next and Last.

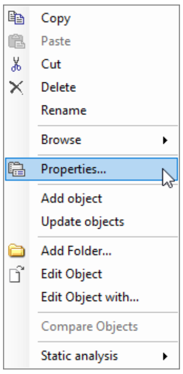
Similar to the Frame visualization also a tab control from Common Controls can be used. The user can also add multiple visualizations to the tab control. For each added visu a tab is created. The Heading as well as an Image can be added for each tab.

6.3 Dialogs

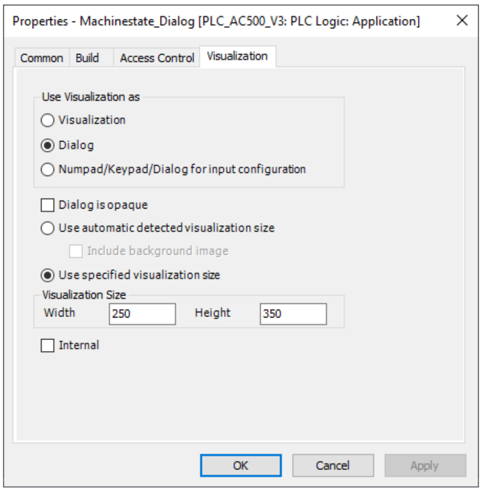
6.3.1 Creation of a Dialog page

Dialogs are normal visualization pages which are used as a pop up window.

After adding a Visualization open the object properties by right click → Properties.

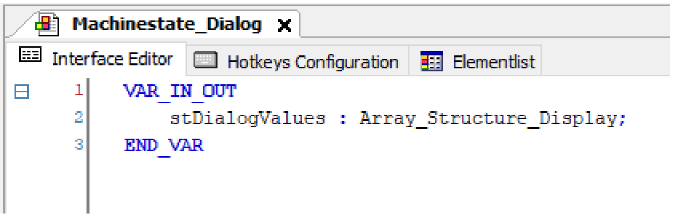


In the tab Visualization the Visualization is used as dialog. Furthermore, the size of this element can be specified.



In case the dialog should not only display a message but also have an input or output, it needs to have an In Out Variable.

In this project the Array variables which are displayed in a Table (see chapter 10.1) can be changed in a dialog page. Each array entry has the type Array_Sructure_Display. Also the dialog Visualization has a variable with this type.



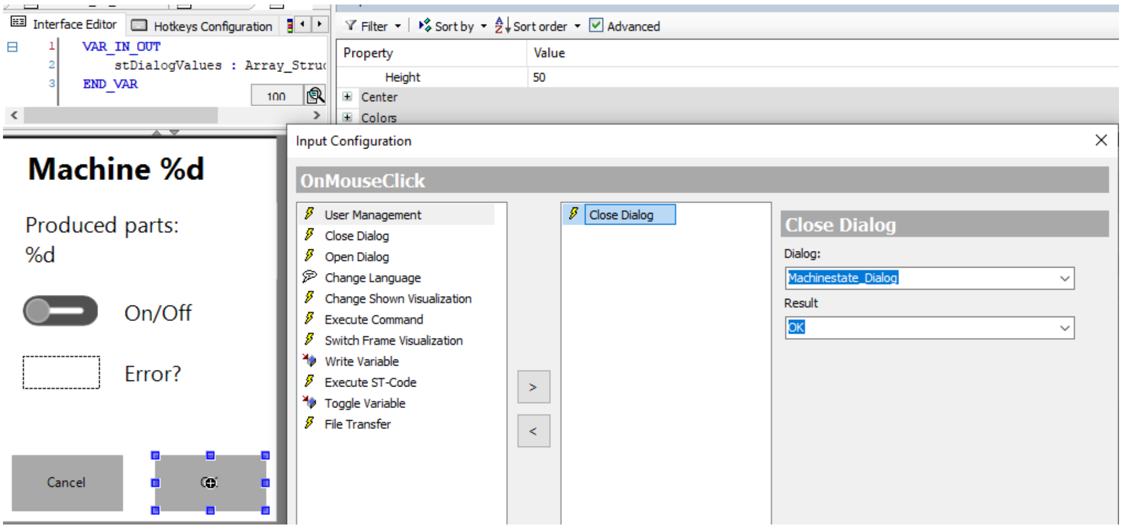
In case the dialog is opened, this local variable is assigned. These values or in case of a struct its entries can be changed in the dialog page. By closing the dialog, the original Project values are overwritten by the local changed values.

In case the project values should be changed immediately the pragma:

{attribute 'VAR_IN_OUT_AS_POINTER'}

can be added in the line above each variable declaration.

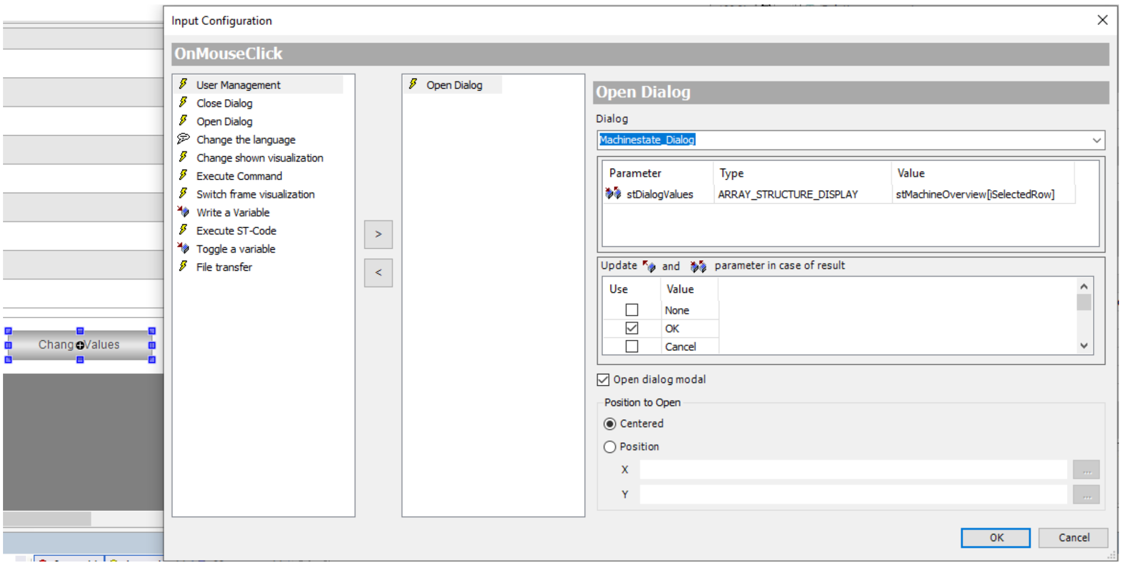
Buttons can be used to close a dialog. In this example the dialog can be closed by an OK and a Cancel button. The input configuration on mouse click is configured. Here Close Dialog is used. As Dialog the self-designed Machinestate_Dialog is used. As Result OK is selected. The configuration of the Cancel button is the same. Only the Result is different.



6.3.2 Calling Dialogs from other visualizations

This dialog should be opened on the page Arrays below the table.

On this page there is a button to change values. Again the On mouse click input configuration is used.



As visible in the picture above the Open Dialog action is used. As dialog the Machinestate Dialog is selected. As this dialog has an In/Out-Variable the structure has to be assigned. Here the Array entry at the selected row is used.

By default, the Position where the Dialog opens is Centered. If the dialog shall be opened next to an element you can specify the position by using the X and Y Position. It is also possible to open the dialog next to the mouse click. Here, ptMouse.iX and ptMouse.iY can be used as position. This is done in the dialog box in the Sandbox.

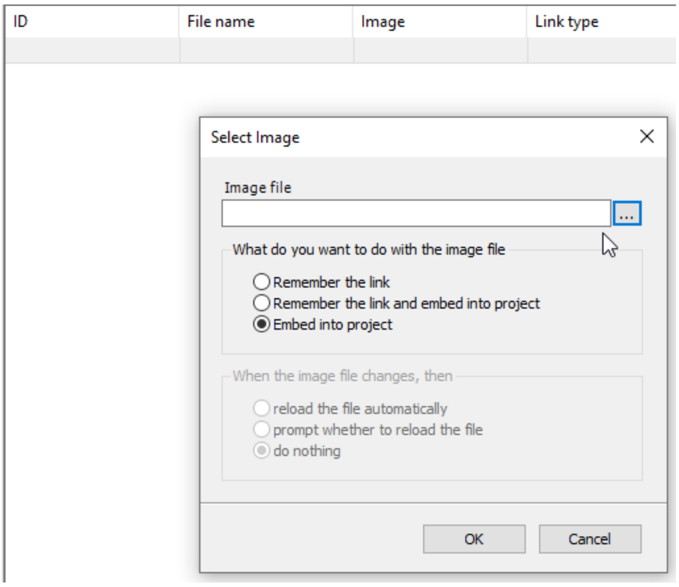
Now it is selectable at which result value the outputs shall be written. It is configured that the Variables are updated by an OK result and not updated by all other results. The names used as value have nothing to do with the behavior. So, you can also specify that the values shall be updated by closing a dialog with cancel. This update is only needed if there are In/Out or Out Variables which are not used as a pointer.

7 Images

To display an image in any visualization an image pool is needed inside the project.

Right click on the Application and add an image pool as object.

After opening the image pool new images can be added either by clicking into Image or via drag and drop.



With the three dots behind Image file an image on the hard disk can be selected.

The image can be included as a link to the picture or embedded into the project.

Remember the link	The picture is not included into the project itself. This means the picture will not be available on other PCs or when the original picture is moved or deleted.
Remember the link and embed into project	The Image is embedded into the project but also the link is remembered. Then you can react in case the image on your hard disk is changed and either do nothing, ask to reload the new image or reload it automatically.
Embed into project	The image will be available also if the original picture is not existing anymore. Changes in the original picture will not be available in the project until new version is uploaded.

Images which are included can be used in different use cases. As visible on the main page AC500_V3_Visu images are used in Image ID to be displayed on a button. In the visualization Language images are added to the page and are configured to change the language.

It is also possible to use an image switcher which can be found in Lamps/Switches/Bitmaps. Here it is possible to change between two images depending on the state of a variable. It is also possible to have a third image which is shown as long as the image is pushed.

All in all, the Image switcher is similar to a Button but able to change the image depending on the variable state. Here the user has the possibility to create his own image switcher to display or change states in the application.

Property	Value
Element name	GenElemInst_33
Type of element	Image switcher
⊟ Position	
X	841
Y	394
Width	150
Height	91
Variable	
⊟ Image settings	
Image on	
Image off	
Image pushed	
Transparent	☐
Transparent color	Black
Isotropic type	Isotropic
Horizontal alignment	Left
Vertical alignment	Top
Element behavior	Image toggler
⊟ Texts	
Tooltip	
⊟ State variables	
Invisible	
Deactivate inputs	
Access rights	Not set. Full rights.

8 Animation

Animations can be used to display progresses, movements or any other kind of changes.

In the section Animation three Pages are used in a tab control.

8.1 Movement Objects

In the Tab move objects two circles are used to show how movements and other animations can be realized. Both circles have the X and Y Position 0. The first circle is hiding the second one.

⊟ Absolute movement	
⊟ Movement	
X	iOffsetX
Y	iOffsetY
Scaling	rZoom
Interior rotation	rRotate

In the properties section Absolute movement variables for the X and Y offset as well as for scaling and Interior rotation are specified.

Depending on the X and Y specified in the movement the circle can be shown on another place of the screen. By increasing or decreasing the assigned variables iOffsetX and iOffsetY inside the project the object is moving.

The Scaling variable is 1000 by default which means as big as drawn in the visu. By changing this number, the object can be drawn bigger or smaller.

There are two circles in the visualization. One is moving automatically. The user has no possibility to change the movement of this object. The other circle can be moved with the arrow buttons and change the size with the + and – buttons.

8.2 Color of objects

By default, two colors of an object can be defined. The normal state and the alarm state. With a Boolean variable the state of this object is changed. In case an object should have more colors or even the user should be able to specify a color a color variable has to be used.

⊟ Color variables	
Toggle color	<toggle/tap variable>
⊟ Normal state	
Frame color	
Fill color	dwColor
⊞ Alarm state	

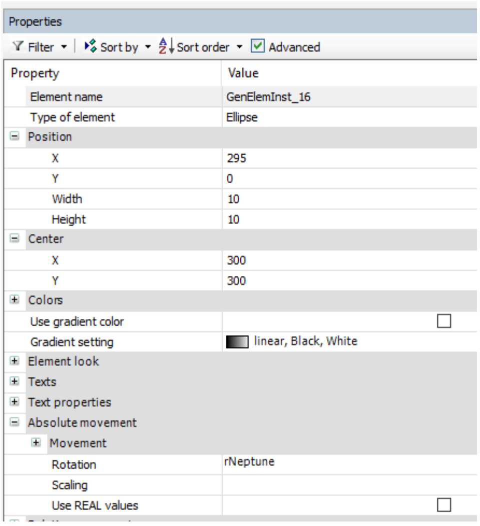
The Fill color is assigned to a variable in the program. A color variable has the type DWORD. The four bytes represent the Transparency, Red, Green and Blue saturation. The value: 0xFFFF00FF displayed in hex means 100 % visible, 100 % Red, 0 % Green and 100 % Blue. This means purple is displayed here. In the project the Visibility is always set to 255 = 0xFF. The Red, Green and Blue values can be changed with sliders in the Visualization.

8.3 Rotate Objects

There are two kind of rotations which are possible. Some objects can only have one of them.

The Interior rotation means the position of the object keeps the same, but it is rotating around its own axis. This is shown in the Move Objects Tab where the user has the possibility to rotate one of the two circles with the two rotate arrows. Similar to the movement the linked variable is increased and decreased as long as the rotate buttons are pressed.

In addition to rotating on its own axis it is also possible to rotate around a fixed point. In this rotation the orientation of the object remains but a movement around one point can be animated. This is done in the tab Solar System. Here the planets of our solar system are shown. They are rotating around the sun. Each planet is a circle. Its orbit is shown with a grey line. To move the planets around the sun on the orbit, they could be moved similar to the circles on the Move Objects visualization, by changing the X and Y Position of the circle. Trigonometric calculations are needed then. In the Visualization this can much easier be realized using the Rotation.



Properties

Filter ▾ | Sort by ▾ Sort order ▾ ☑ Advanced

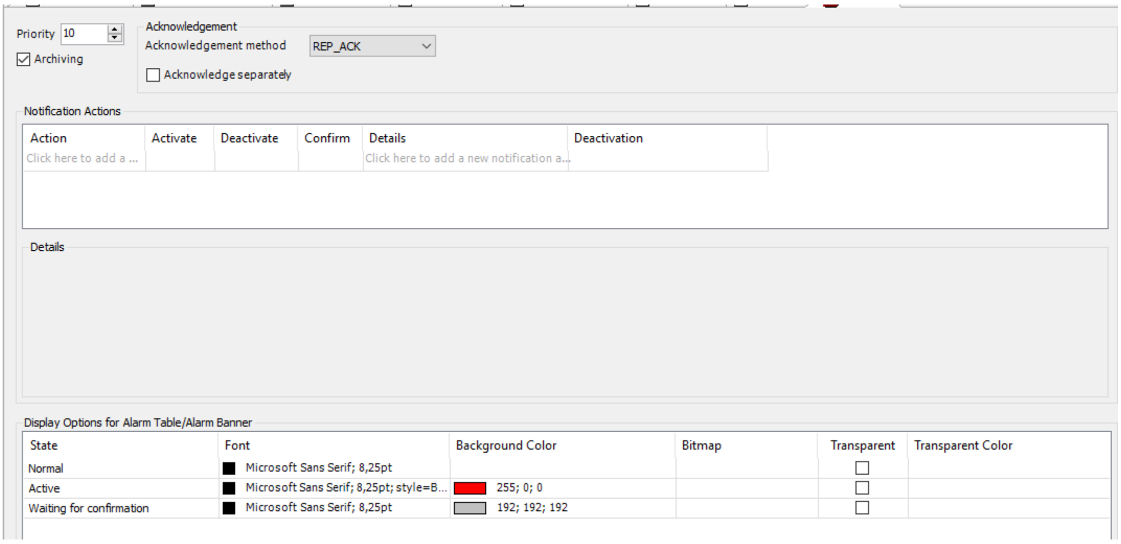
Property	Value
Element name	GenElemInst_16
Type of element	Ellipse
⊟ Position	
X	295
Y	0
Width	10
Height	10
⊟ Center	
X	300
Y	300
⊞ Colors	
Use gradient color	☐
Gradient setting	linear, Black, White
⊞ Element look	
⊞ Texts	
⊞ Text properties	
⊟ Absolute movement	
⊞ Movement	
Rotation	rNeptune
Scaling	
Use REAL values	☐

As visible in the screenshot above the circle has a fixed position at 10, 10. But the center of the object is at 300, 300. With the Rotation an object can be rotated around its Center. In this example the Circle which is representing the planet Neptune is rotating around 300, 300 which is the center of the sun. The variable rNeptune gives the position in degrees. In the program rNeptune is increased each cycle. In the visualization the Neptun cycle rotates around the sun. The same is done for all other Planets. As the rotation variable is increased with different rates some planets are rotating faster and some slower.

9 Alarm Management

To have an Alarm Management in the visualization the Alarm Configuration has to be added to the Application first. By default, the Alarm Configuration has three types. Error, Info and Warning.

The configuration for errors is described here. It is similar for Warnings and Infos.



Each class has a priority. The default priority of errors is 10. Below, with the checkbox “Archiving”, it can be selected if errors should be archived to have them visible in an alarm history.

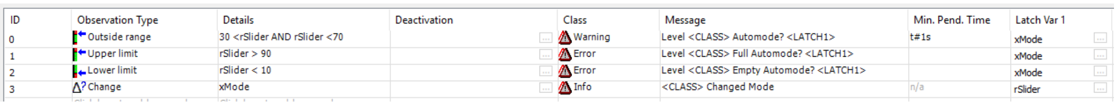
In the group Acknowledgement it is possible to select how errors should be acknowledged.

In the group Display Options for Alarm Table/Alarm Banner the colors of the Errors shown in a visualization can be configured. Here all states of the Alarms can be specified.

Similar to the Errors also Warnings and Infos can be configured. It is also possible to add own alarm classes to have a more detailed reaction depending on the kind of alarms.

An Alarm Group needs to be added to the alarm configuration. Depending on the project there can also be more than one alarm group added to separate between different parts of the system.

In this program only simple alarms are configured. A slider is representing a fill level in the Range between 0 and 100. In each line of the Alarm Group new Alarms can be added. In this example there are four alarms configured.



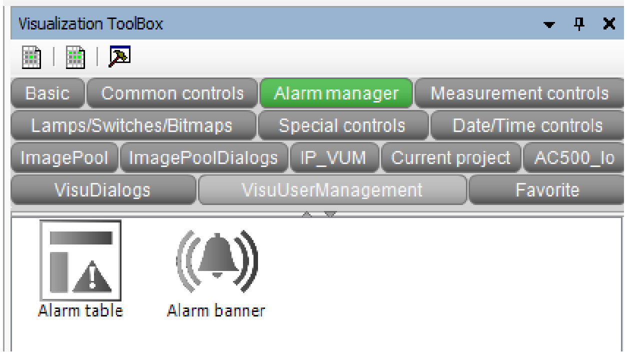
In Observation Type it can be specified which behavior should cause an alarm. In the first line Outside range is set. In the details the Variable to monitor as well as the Borders if necessary are set. Here it is between 30 and 70. The alarm class is a Warning. In the Message the displayed Text is shown. Here it is Level <CLASS> Automode? <LATCH1>. In the pointed brackets Placeholders are used. So if the Waring is shown later in a Visualization it will be displayed: “Level Warning Automode? FALSE”. The min Pend. Time is the time the alarm has minimum to be active until it is shown in the visu or further actions if configured are executed. Here it is one second. If the slider value goes up to 80 but returns to a value below 70 within one second, no warning will be thrown.

In the Latch Variables other variables can be added. These Variables are not monitored to throw an alarm but might be useful to know when analyzing the alarm history. In this example it might make a difference if an alarm is thrown in the automatic modus or in the manual modus.

Similar to this Warning two Alarms are configured to check if the value is above 90 or below 10. The last entry in this list is a change. Each time the variable xMode is changed an info is shown. This way it can be logged when the machine was changed from Manual to Auto mode or back.

The alarm Message is stored in a Text list named the same name as the alarm group.

In the Visualization the user has two objects in the Alarm manager. An Alarm table and an Alarm banner.

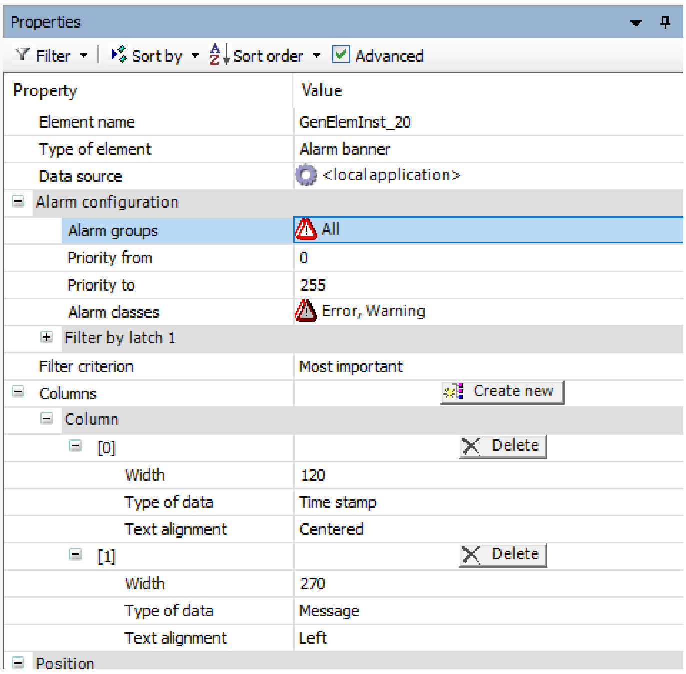


9.1 Alarm Banner

In the properties of the alarm banner the alarm configuration and filtering can be done.

If more than one alarm group is added in the project, selected groups can be displayed in the banner. As in this program only one alarm group is used All is selected. Furthermore, it can be specified that only alarms in a priority range are shown.

As there are different alarm classes, it is also possible to select the classes which should be shown in the banner. Error and Warning is selected. That also means Infos are not shown in the alarm banner.

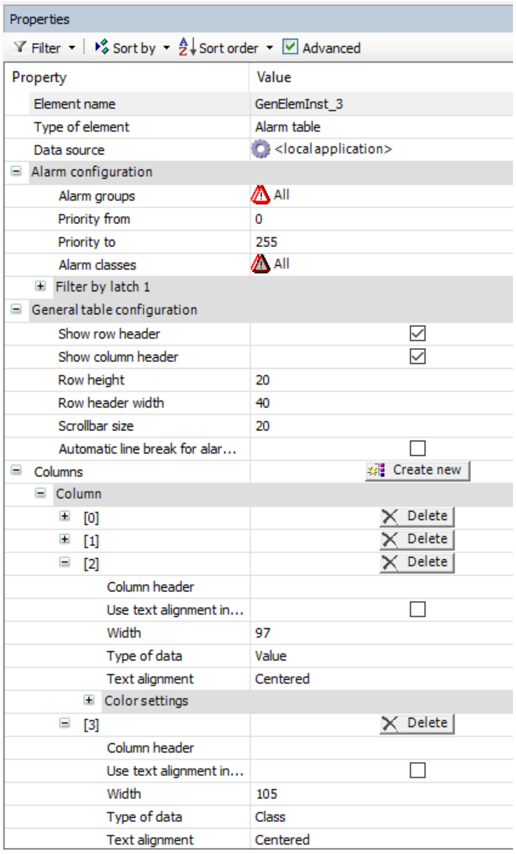


With the Filter it can be selected if the most important or the most recent alarm should be shown in the banner.

In the section columns the user can specify which columns should be shown. In this alarm banner only the timestamp and the message are shown.

9.2 Alarm Table

The configuration of the alarm table is similar to the Alarm Banner. Only differences are described in this chapter.



Properties

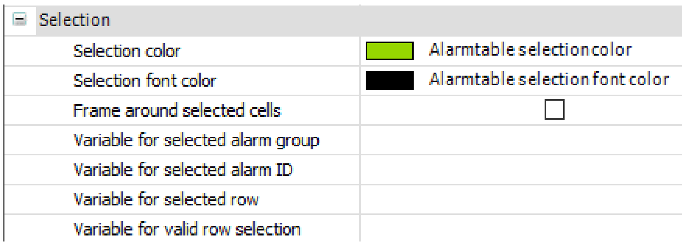
Filter | Sort by | Sort order | Advanced

Property	Value
Element name	GenElemInst_3
Type of element	Alarm table
Data source	<local application>
Alarm configuration	
Alarm groups	All
Priority from	0
Priority to	255
Alarm classes	All
Filter by latch 1	
General table configuration	
Show row header	☑
Show column header	☑
Row height	20
Row header width	40
Scrollbar size	20
Automatic line break for alar...	☐
Columns	Create new
Column	
[0]	Delete
[1]	Delete
[2]	Delete
Column header	
Use text alignment in...	☐
Width	97
Type of data	Value
Text alignment	Centered
Color settings	
[3]	Delete
Column header	
Use text alignment in...	☐
Width	105
Type of data	Class
Text alignment	Centered

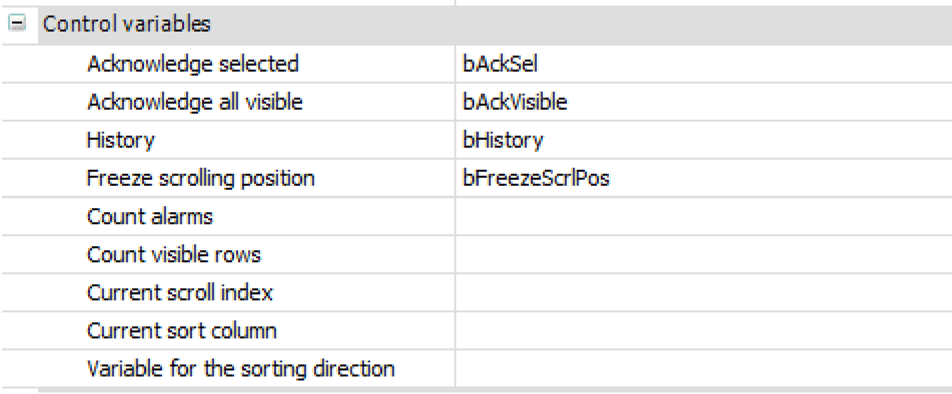
The alarm configuration and the columns which are already described in the last chapter are nearly the same in this project for the Alarm Table. Here All alarm classes are shown. In addition to the Alarm timestamp and message also the value and the alarm class is displayed.

In the alarm table additional settings are possible. In the section general table configuration, the user can specify if the headers should be shown and which size the columns should have. Furthermore, the scrollbar size as well as an automatic line break can be specified.

It is possible to select a row in the alarm table. Details on the selected alarm can be shown. Therefore, the variables in the section Selection can be used. In this project no variables are configured.

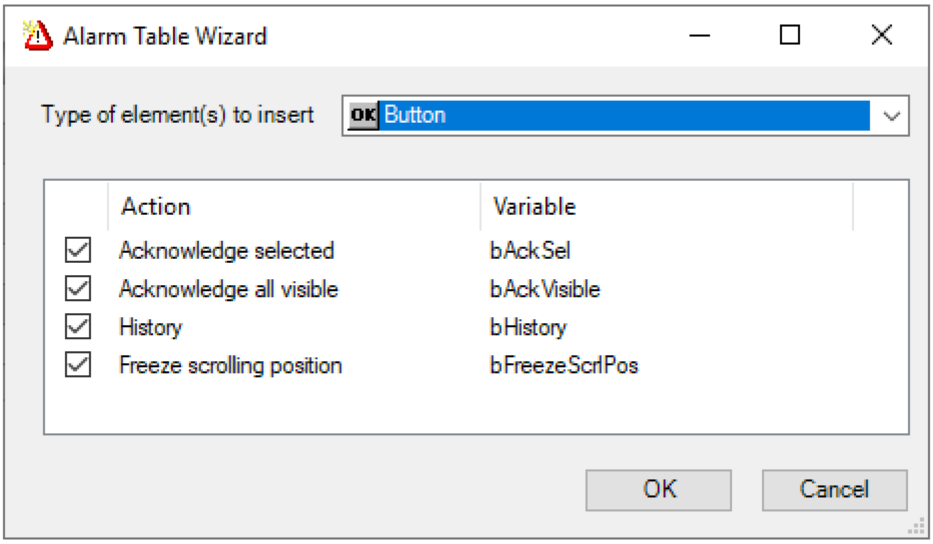


Selection	
Selection color	Alarmtable selection color
Selection font color	Alarmtable selection font color
Frame around selected cells	☐
Variable for selected alarm group	
Variable for selected alarm ID	
Variable for selected row	
Variable for valid row selection	



Control variables	
Acknowledge selected	bAckSel
Acknowledge all visible	bAckVisible
History	bHistory
Freeze scrolling position	bFreezeScrlPos
Count alarms	
Count visible rows	
Current scroll index	
Current sort column	
Variable for the sorting direction	

In the section Control variables, it is possible to have variables to count the alarms or visible rows as well as the scroll index, the column which is used to sort the alarms and the sorting direction. These variables are not used. Only the buttons to acknowledge the selected alarm or all visible alarms as well as the buttons for alarm history and freezing the scrolling position are used. These buttons can be generated automatically. Therefore, right click on the alarm table in the visualization and select: *Insert Elements for acknowledging alarms…*



Select the Buttons you want to use in your visualization and continue with OK. Afterwards the buttons are added automatically below the Alarm table.

10 Displaying arrays

Arrays are frequently used in applications. To display the content of an array two possibilities are shown in this application. On the one hand the content is displayed in a table and on the other hand the data is plot into a X/Y diagram.

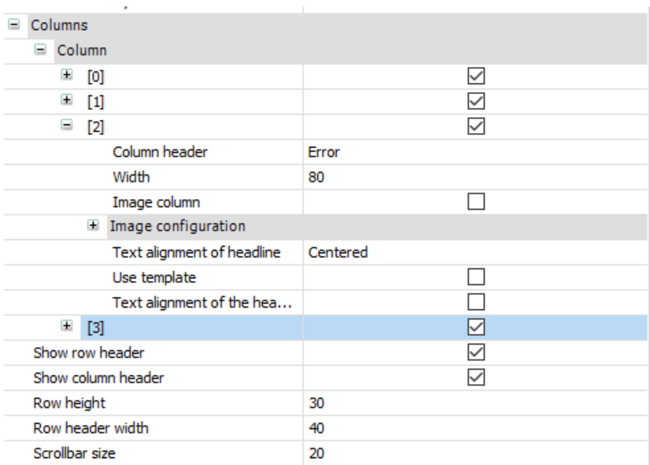
10.1 In a Table

As an example, a machine overview should be displayed in a table. 20 machines are numbered, have two flags to highlight if they are running or have an error. Furthermore, an integer value which displays the produced parts up to now.

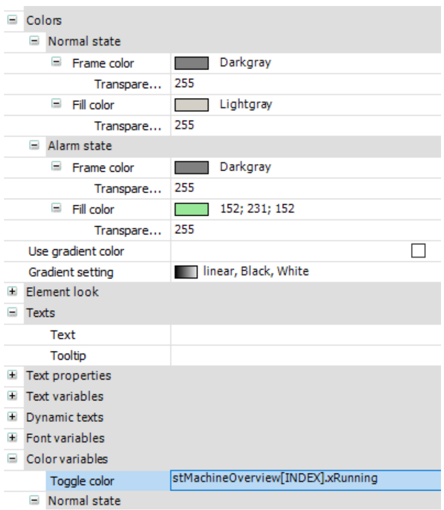
The information above should be displayed in a table. Therefore, a table which is part of the Common controls is added to the visualization. In its properties the Data array which should be displayed is linked.

For each Variable in the structure a column in the table is created. With the check box behind it can be selected which information should be shown.

It is possible to configure each row as well as the headers and the scrollbar in this section.

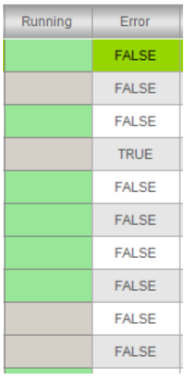


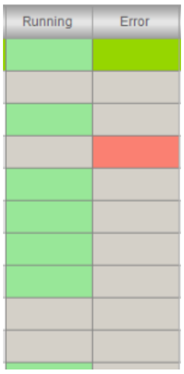
In case further settings needs to be done for this column the checkbox Use template can be checked. Then the color of the column, the displayed text format or an input configuration for changing the values can be configured.



In this example, the columns Running and Error were modified. By default, the columns show a string value which is either TRUE or FALSE. To have a better overview running machine cells are highlighted green if the machine is running and grey if it is off. The text value TRUE or FALSE is not shown anymore. The same is also done for the Error column. Here an error is highlighted in red.

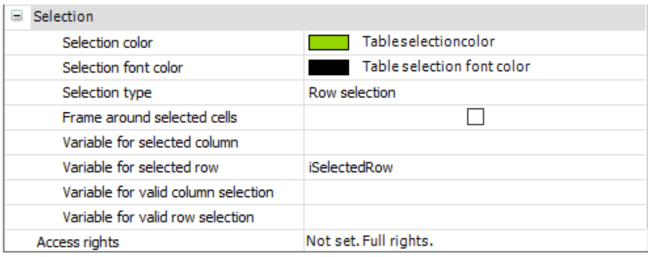
Following pictures show the contrast, on the right side the modified columns and on the left side the columns where only Running was modified.





By modifying the columns as described you might not be able to see the selected row/cell in this column and you lose other settings like highlighting each second line.

A highlighting like this could be manually programmed by using different colors depending on the row but these modifications are complex and time consuming.



Selection	
Selection color	Tableselectioncolor
Selection font color	Table selection font color
Selection type	Row selection
Frame around selected cells	☐
Variable for selected column	
Variable for selected row	iSelectedRow
Variable for valid column selection	
Variable for valid row selection	
Access rights	Not set. Full rights.

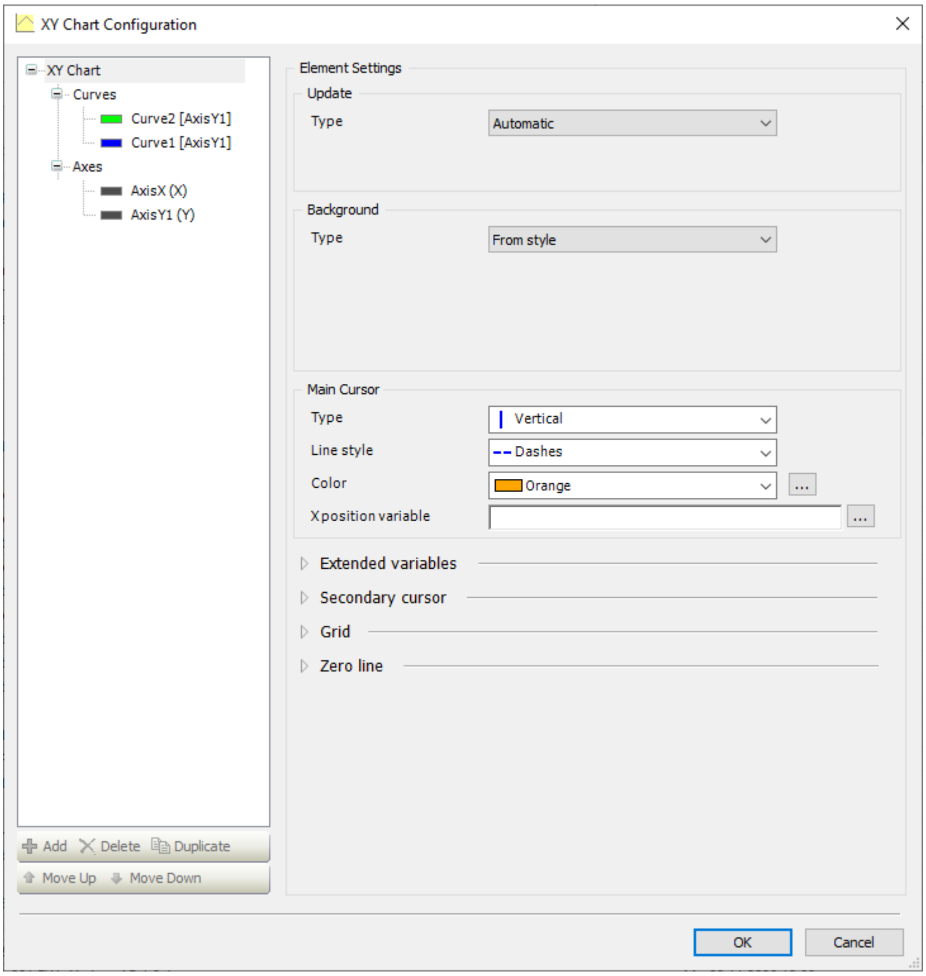
In the properties selection it can be configured whether a cell, row, column, row & column or nothing can be selected. It is furthermore possible to link the selected column or row to any variable.

In this example the values in the selected row can be changed by clicking the Change Values button. Then a dialog window opens, and the user can change the values. The dialog settings are described in chapter 6.3.

10.2 In a XY-Plot

In some cases it is required to visualize an array as a XY plot, e.g. raw data of an analog input. The diagram of the XY plot gives a better overview such data in comparison to a table view. A plot of this data in a diagram is needed to for example check an analog input signal or monitor other conditions of your system.

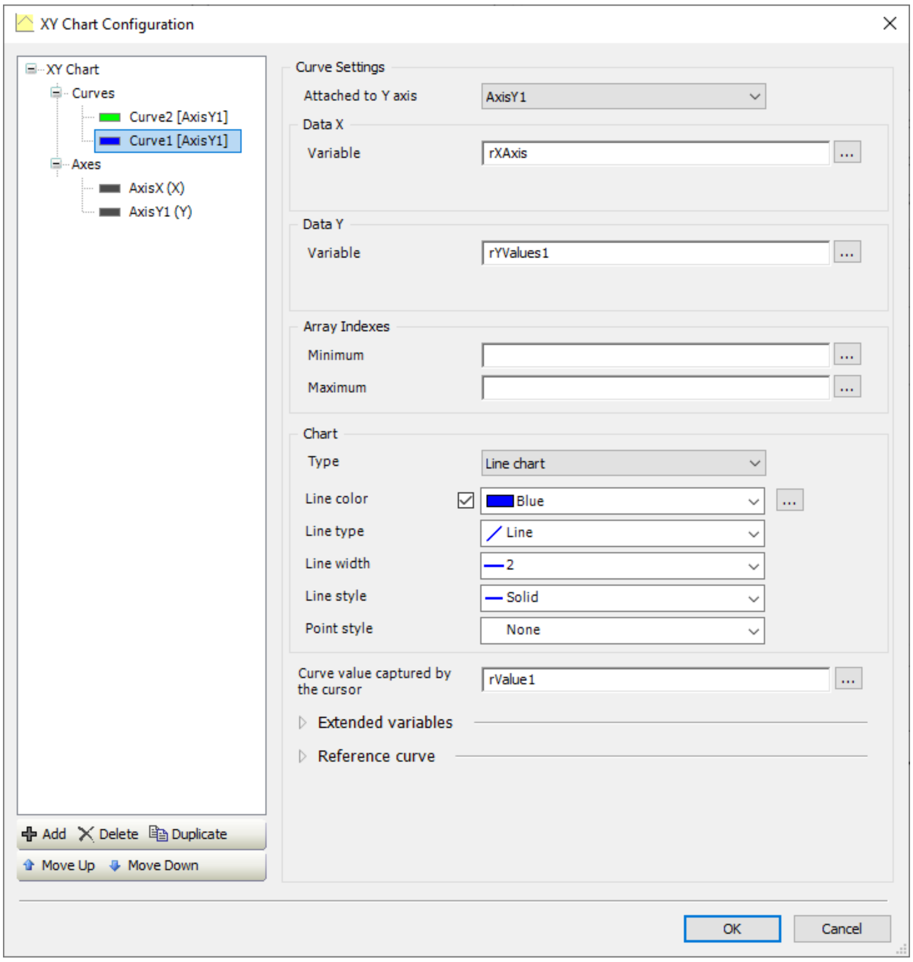
To plot the data the Cartesian XY chart from the Special controls can be used. The configuration of Cartesian XY chart as well as the curves and the axes can be done.



In the chart configuration the cursor can be enabled. A Curve can be added by right clicking on Curves and select add.

For each curve the Y Axis can be specified. The cart can have more than one Y Axis. In Data X and Data Y the Arrays which should be plotted are entered. In case not the complete arrays should be plotted the Min and Max Array indexes can be specified. In the chart configuration the Line Properties like Type, Color, Style and Width can be changed.

Also the usage of a cursor is possible, which can be linked to a variable, which can be used to get the values in the plot.



To zoom inside chart, the zoom functionality must be enabled in the section Control variables. Further configuration settings for zoom are available to undo the last zoom change or to reset the zoom to home where the complete plot is shown again. Same settings are also possible for pan. Please make sure that you are using a multitouch display when working with these settings.

Control variables	
Zoom	
Enable	xZoomEnable
Home	xZoomHome
Undo	xZoomUndo
Is zoomed	xIsZoomed
Pan	
Enable	xZoomEnable
Home	xZoomHome
Is panned	

Note: To move and use the cursor inside the diagram, zoom and pan must be deactivated.

11 File Handling

Text files can be created, viewed, edited and saved with a File Editor. Furthermore, it is possible to search for a content inside the file. If the text editor from Special controls is added to a visualization its properties can be edited.

The File variables contains the path and name of the used text file and flags to open, close, save and create a file. The Edit variables also contains a string with the text to search for as well as flags to find this string and find the next occurrence.

There are also variables for the cursor position and the selected text. But these are not used in the project. In the section error handling a variable for the error codes as well as flags for content change and edit mode can be linked.

In new files it can be configured how new files should be created. By default, it is an ASCII file with carriage return and line feed for each new line. These settings can also be changed.

⊟ Control variables	
⊟ File	
Variable	sFileName
Open	xFileOpen
Close	xFileClose
Save	xFileSave
New	xFileNew
⊟ Edit	
Variable	sFileSearch
Find	xFileFind
Find next	xFileFindNext
⊟ Cursor position	
Line	
Column	
Position	
Set trigger	
⊟ Selection	
Start position	
End position	
Start line number	
Start column index	
End line number	
End column index	
Line to select	
Set selection	
⊟ Error handling	
Variable for error code	usiFileErno
Variable for content changed	xFileContentChanged
Variable for access mode	xEditMode
Maximum line length	200
Editor mode	Read/Write
⊟ New files	
Encoding	ASCII
New line character sequence	CR/LF
Access rights	Not set. Full rights.

The two string variables for filename and search text are displayed in a textbox and can be changed by the user. The flags for the file and edit operation are toggled by buttons. Once the operation is done the file editor resets the linked variable back.

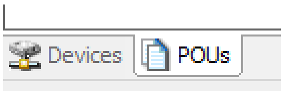
The error number is displayed as String in a textbox and the content changed message is only shown when the flag is true. The edit mode variable is not used as the text editor is always in Read/Write mode.

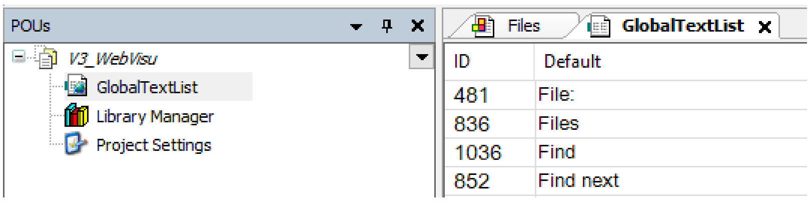
12 Multilanguage

12.1 Add Languages and Translate Visualization Content

> Note: The Multilanguage configuration should be done once the rest of the Project is finished.

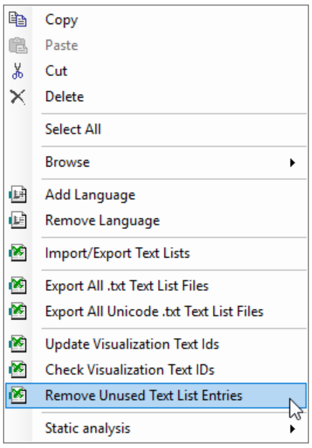
To add languages, change to the tab POUs and open the file GlobalTextLists.





Here you will find all entered texts with an ID and the Default string.

Even if you add a textbox, change the content and delete the textbox the name will be visible in the GlobalTextList. You can first reduce the shown Strings to the once which are really used in the project. To do so, right click in the GlobalTextList and select Remove Unused Text List Entries.

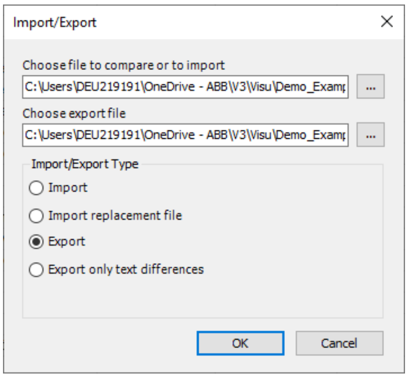


Once this is done your Text List is reduced to the used Entries.

With a right click you can select Add Language. This way you can add multiple columns for different Languages to the Global Text List.

To configure the Multilanguage, you can now translate each entry into all languages inside Automation Builder. It is more comfortable to export the list as csv file. In this example it is shown how to work with a CSV.

Please create somewhere an empty csv file. Go back to the Global Text List and select Import/Export Text Lists. Specify the Export file, select Export and confirm with OK.



This export file can then edited with a text editor, for example with Excel. The different languages can be found in different Columns.

	A	B	C	D	E	F	G	H	I	J	K
1	TextList	Id	Default	English	Italian	German	French	Chinese	Greek	Spanish	Russian
2	GlobalTextList		%1.0f								
3	GlobalTextList		%1.1f								
4	GlobalTextList		%1.2f								
5	GlobalTextList		%d								
6	GlobalTextList		%s								

The advantage of the CSV is that you have all text lists in one file. There is not only the Global Text List inside but also all text lists for example from the Alarm Configuration, the Diagnosis or self-designed messages.

This document can either be translated by yourself, given to any person who will translate it into another language or translate it by an online translator.

In this example the document was translated with an online translator. Please be careful that translations might be incorrect. Please make sure by auto translation that string constants like %d or RN might be also translated which is not wanted.

Note: Auto translation might also translate string constants like %d or RN. This is not wanted and should be checked manually after the translation.

Furthermore, please make sure that the default CSV file is in ASCII format which will not be able to show any special characters like for example used for Chinese language.

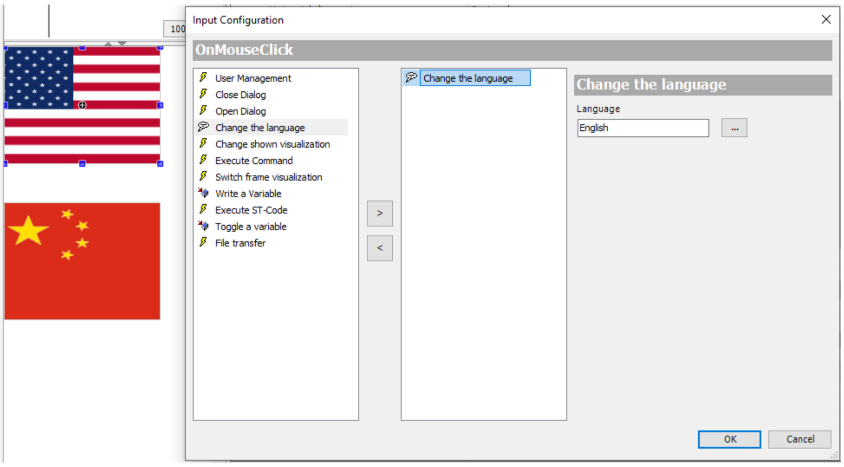
Once the document is translated you can right click into the text list again, import/export a file and select the new CSV file for importing.

Files | GlobalTextList

ID	Default	English	Italian	German	French	Chinese	Greek	Spanish	Russian
885	%1.0f	%1.0f	%1.0f	%1.0f	%1.0f	%1.0f	%1.0f	%1.0f	%1.0f
16	%1.1f	%1.1f	%1.1f	%1.1f	%1.1f	%1.1f	%1.1f	%1.1f	%1.1f
711	%1.2f	%1.2f	%1.2f	%1.2f	%1.2f	%1.2f	%1.2f	%1.2f	%1.2f
502	%d	%d	%d	%d	%d	%d	%d	%d	%d
312	%s	%s	%s	%s	%s	%s	%s	%s	%s
939	+	+	+	+	+	+	+	+	+
455	-	-	-	-	-	-	-	-	-
434	1234	1234	1234	1234	1234	1234	1234	1234	1234
766	AC500 V3 Visualizati...	AC500 V3 ...	Esempio di ...	AC500 V3 Visuali...	Exemple de ...	AC500 V3可...	Παράδειγ...	Ejemplo de visu...	Пример визу...
430	Access Right	Access Right	Diritto di acc...	Zugangsrecht	Droit d'accès	访问权	Δικαίωμα ...	Derecho de acc...	Право досту...
414	ACK all visible	ACK all visi...	ACK tutto vi...	ACK alle sichtbar	ACK tout visi...	ACK所有可见	ΖΗΤΗΣΤΕ...	ACK todo visible	ACK все вид...
413	ACK selected	ACK selected	ACK selezio...	ACK ausgewählt	ACK sélectio...	选定ACK	Επιλέχθηκ...	ACK seleccionado	выбранная ...
198	Admin	Admin	Admin	Verwaltung	Admin	管理员	διαχειριστής	Admin	Администра...
1049	Alarm	Alarm	Allarme	Alarm	Alarme	报警器	Τρομάζω	Alarma	Сигнал трев...

12.2 Change the Language from the Visualization

In the current state all used texts are translated in the different languages. To change the languages any object can be used. In this example a Frame with a Picture of a flag is used but you can also use a normal button. In the Input configuration the on mouse click event *Change the language* is added and assigned to any project language. Here The language is changed to English.

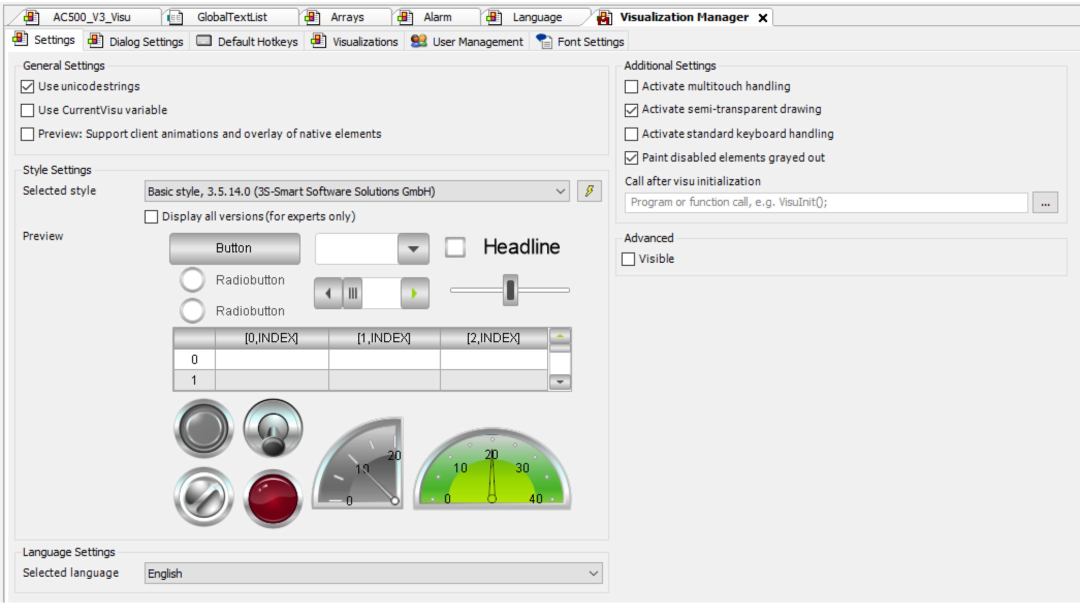


This way all other buttons for other languages can be added.

	Note: Changing a language is a global setting. If multiple clients are connected to the visualization the language is changed for all connected clients.

12.3 Settings in Visualization Manager

In the Visualization Manager two settings regarding the Multilanguage can be made.



In the Settings the Selected language can be chosen. The visualization starts with this language. If not specified default is taken.

In the tab Font Settings, the user can change the Font sizes factor for each language to fit the font size if a language needs more/longer words than others.

It is also possible to reduce the size automatically to the size of the text box.

13 Recipes

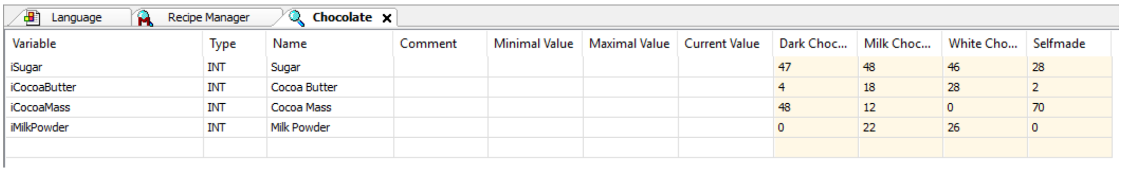
13.1 Recipe Manager

To use recipes in a visualization a Recipe Manager has to be added as object to the application. Right click to the Recipe Manager and add a new Recipe Definition. In this project it is called Chocolate.

In *Variable* the project variables which shall be changed according to the recipe are linked. In Name you specify the Name for this element.

By right clicking a new Recipe can be added. In this project the four Recipes *Dark Chocolate, Milk Chocolate, White Chocolate* and *Selfmade* are added.

For each recipe the values of the different products can be specified. At the end the table might look like this.

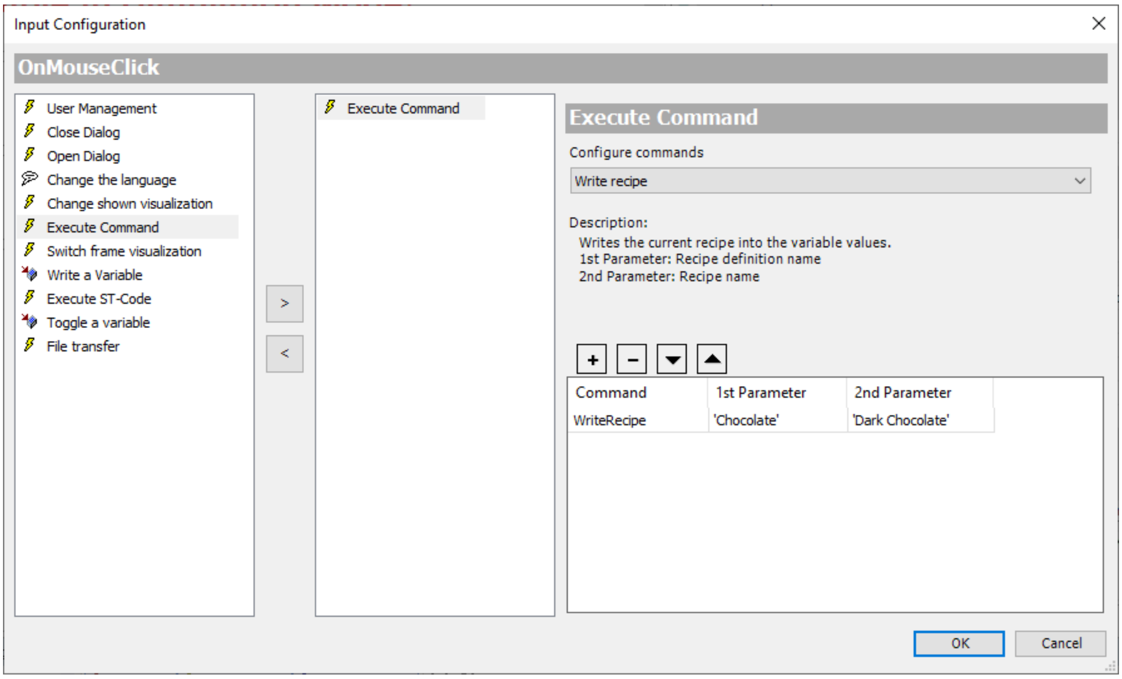


Variable	Type	Name	Comment	Minimal Value	Maximal Value	Current Value	Dark Choc...	Milk Choc...	White Cho...	Selfmade
iSugar	INT	Sugar					47	48	46	28
iCocoaButter	INT	Cocoa Butter					4	18	28	2
iCocoaMass	INT	Cocoa Mass					48	12	0	70
iMilkPowder	INT	Milk Powder					0	22	26	0

13.2 Settings in Visualization

The products Sugar, Cocoa Butter, Cocoa Mass and Milk Powder are shown in the visualization as slider values. To change between the different Recipes buttons are used.

In the input configuration of the button on mouse click event *Execute Command* is used.



In the drop-down menu Configure commands Write recipe is selected and added with the + button. As 1st Parameter the Recipe Definition here *Chocolate* is entered as string. The 2nd Parameter is the Recipes Name here for example *Dark Chocolate*.

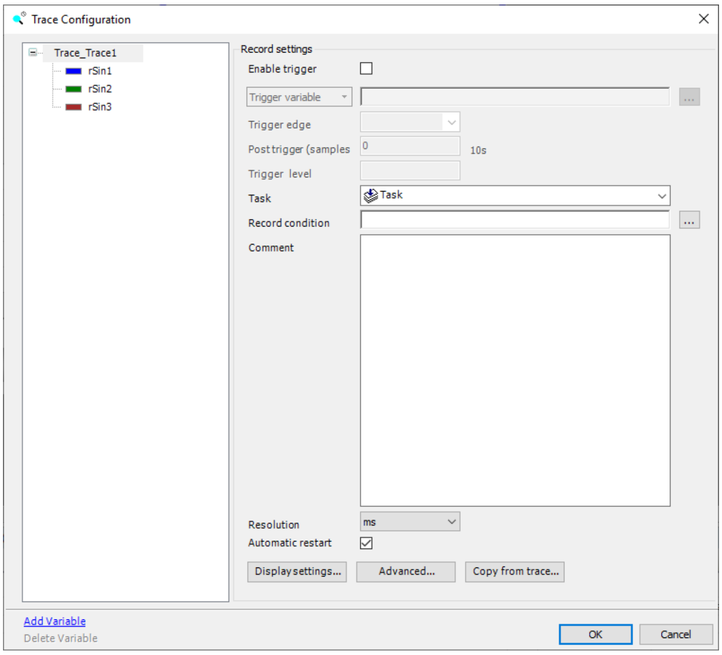
Like this, buttons for all other recipes are designed. To store the current slider position in a recipe the command Read recipe is used in the same way.

14 Traces

A trace can be used to show live values on the screen.

A Trigger to start the trace can be configured, as well as the number of samples which shall be recorded after the trigger is activated.

To the Trace itself multiple variables can be added. For each Variable the line and point style as well as warning levels and a variable for the visibility can be set.



In the properties of the Trace variable for starting and stopping the trace can be set. Also a cursor can be configured which shows values when the trace is stopped.

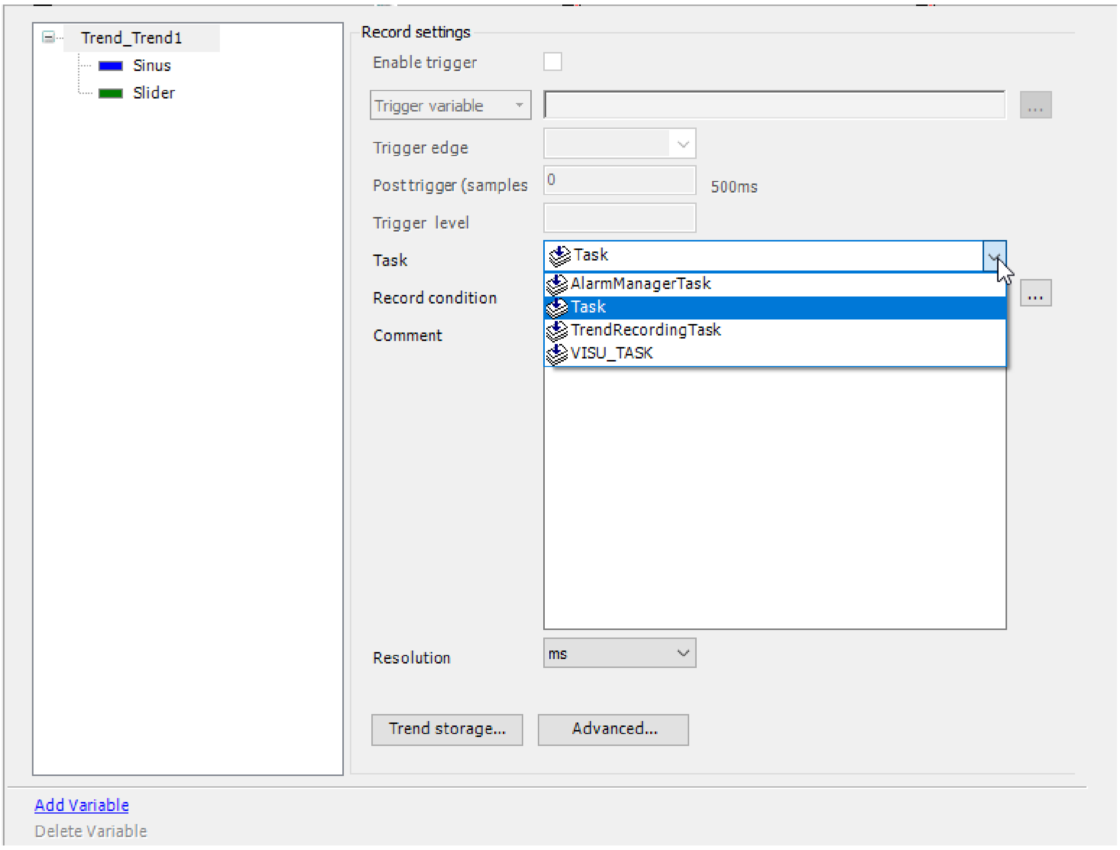
In contrast to a Trend a Trace only stores the last 1000 values. The Trace is organized in a ring buffer which overwrites the oldest values. No history is available for a trace. The size of the buffer as well as the sampling time can be configured in the advanced settings

15 Trends

A Trend from the section Special controls can be added to a visualization.

At first the Task which is recording the trend has to be selected. A Trend Recording Task is automatically added to the task configuration. This task handles the trend data and the update of the values but for recording the Application Main Task should be chosen.

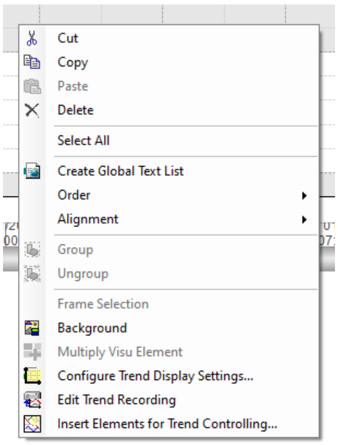
In the trend storage it can be selected what the maximum size of the trend file is. By default it is 16 MB. In the advanced settings the user can select if each cycle a trend value shall be sampled or only each 2, 3 or 10 cycles.



The Trend Recording Manager is automatically added to the application and the trend is added as object. By opening Trend_Trend1 it is possible to change the trend settings or variables.

Variables can be added to the trend by selecting Add Variable. A variable from the program is added which is a sinus oscillating as a sinus wave. In addition, a Slider Value is added which can be changed by the user in the visualization.

By default, the last 10 seconds of the Trend are visible. By right clicking on the trend Insert Elements for Trend Controlling can be chosen. Here the Legend, Time range picker and date range picker can be added automatically. In some Automation Builder versions before AB2.4.0 this context menu is not available then the user needs to add the elements manually and link to the Trend.



In the properties of the trend a cursor can be configured. The cursor is only shown when the trend is not running. So the Date Range picker, see chapter 15.2, is not in the last position.

The format for the X Axis Legend can be specified.

Show cursor	☑
Show tooltip	☑
Show frame	☐
Number format	
⊟ Tick mark labels	
Time stamps	Absolute time stamps
Draw labels on two lines	☐
Omit irrelevant information in timesta...	☐
⊟ Internationalization (formatstrings)	
Date	
Time	HH:mm:ss

15.1 Legend

The Legend element is part of the special controls. It can be added to a Trend. If a legend and a Trend is on the same visualization page, they can be linked with each other. In the Trend the Legend Element is attached as control element. This can be done automatically since AB2.4.0

⊟ Attached control el...	
Date range se...	GenElemInst_5
Time selector	GenElemInst_25
Legend	GenElemInst_7

The Legend is attached to the trend element.

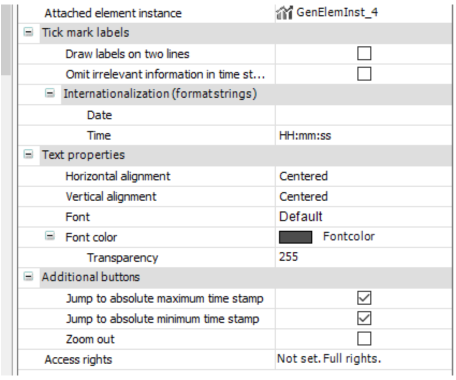
Orientation	Horizontal
Attached element ...	GenElemInst_4
⊟ Layout	
Maximum num...	1
Maximum num...	2

In the Layout of the legend the maximum number of rows and columns can be set. This way the legend can either be in a row or in a column.

When being logged in or in a web visualization the legend contains the variable names from the trend as well as the configured columns and the line/ dot styles.

15.2 Date Range Picker

The date Range picker is part of the Date/Time controls. Its linkage is the same as the Legend.



The format for the Date and Time displays can be specified. Buttons for jumping to the Start or End can be activated. Also zooming out to show the complete recording.

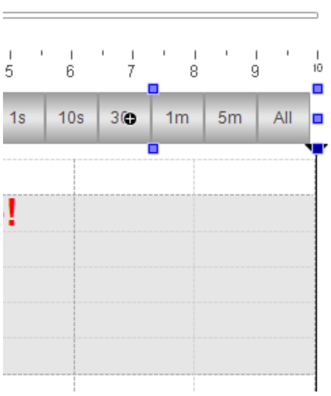
If the position of the date range picker is changed the trend stops moving and if a cursor is configured it becomes visible. To see the live trend again the picker is moved to the right and started again.

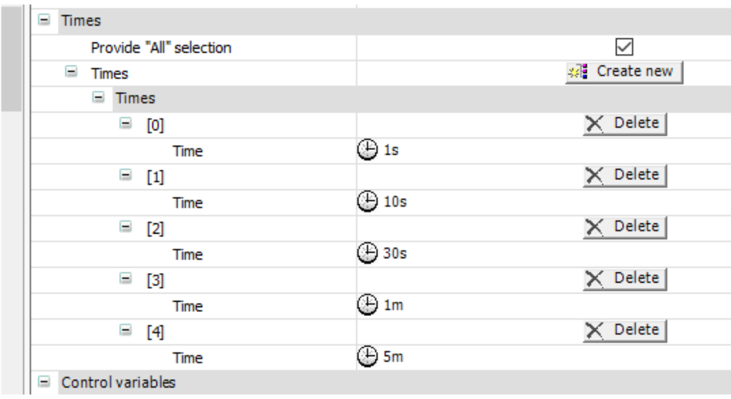
15.3 Time Range Picker

The Time Range Picker is also an element which can be linked to the trend similar to the Legend. By default, the last 10 seconds are shown in the trend. With a Date Range Picker it is possible to change the shown time by increasing or decreasing the size of the scrollbar.

With a Time Range Picker it is easily possible to show a specified time like 10 seconds or one minute in the trend window.

In the configuration Times can be added to the time range picker. In this case 1/10/30 seconds and 1/5 minutes have been added. As shown in the visualization these settings are then visible in a list. The *All* selection can also be added to the time range picker as an additional button. This is shown in the screenshot below but not used in the project.





In the running visualization the used time is highlighted. Other times can be chosen by selecting another time button.

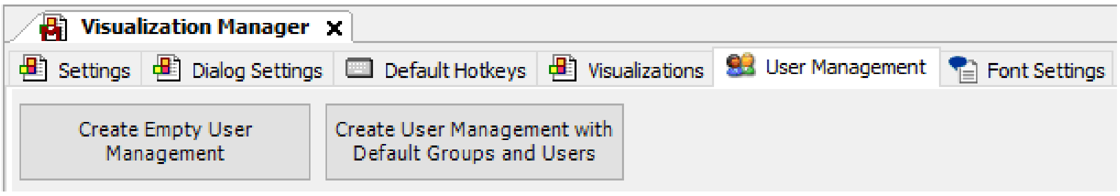
16 User Management

In this chapter the settings in the visualization are described.

For more information and about user management in AC500 V3 please check Application Note AC500 User management with V3 3ADR010315 which is available in the ABB library.

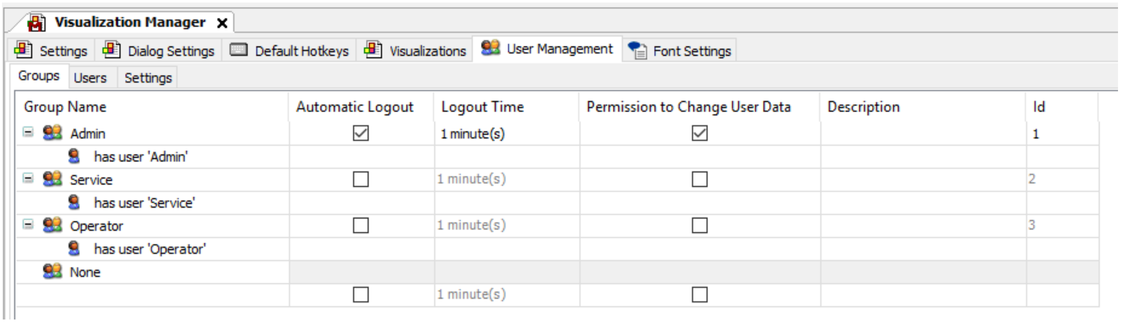
16.1 Visualization Manager Configuration

The user management for the visualization can be activated in the Visualization Manager.



Here in the tab User Management you can either create an empty User Management or Create a User Management with Default Groups and Users.

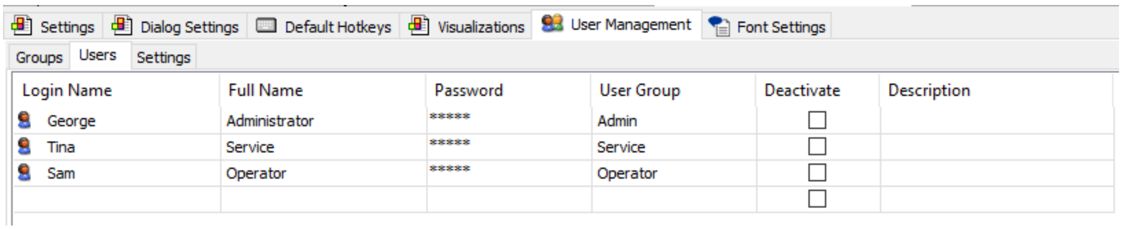
By selecting the default groups Admin, Service and Operator are added to the User management. Each group has a user called Admin, Service or Operator.



Group Name	Automatic Logout	Logout Time	Permission to Change User Data	Description	Id
Admin	☑	1 minute(s)	☑		1
has user 'Admin'					
Service	☐	1 minute(s)	☐		2
has user 'Service'					
Operator	☐	1 minute(s)	☐		3
has user 'Operator'					
None					
	☐	1 minute(s)	☐		

Each group can be configured. The Administrator is the only group which has the permission to change the User Data. The Administrator is also logged out automatically after one minute. In the example all user groups are logged out after one minute.

In the subtab *Users* the Login credentials for all users can be changed also further users can be added. In this Example the users Admin, Service and Operator are renamed to "real" persons. George is an admin, Tina Service and Sam Operator.

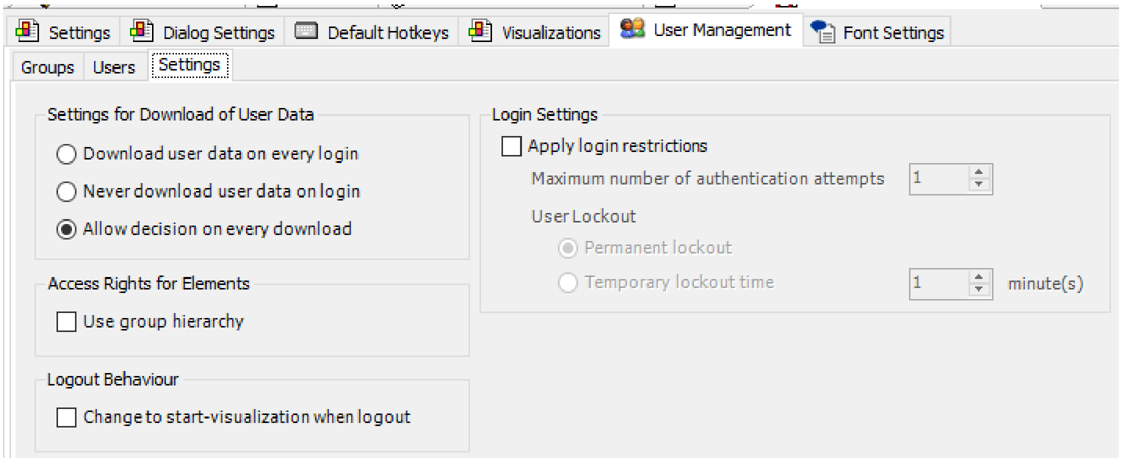


Login Name	Full Name	Password	User Group	Deactivate	Description
George	Administrator	*****	Admin	☐	
Tina	Service	*****	Service	☐	
Sam	Operator	*****	Operator	☐	
				☐	

The password for each user is specified here. All have 1234 in this project.

In the tab *Settings* more detailed configuration can be done.

As described multiple users can be added to the visualization user management in the users tab. Users can be deactivated or passwords can be changed. In the running system the administrator has the possibility to change the user management configuration. The users can only change their passwords when being logged in. In the settings it can be defined whether the user management shall be downloaded or not. With the three possibilities the user data is downloaded on every login which overwrites the data inside the PLC, never which keeps the user data as it is or as decision on each download.



A user group hierarchy can be activated which means any user has automatically all rights which also a user in the hierarchy below has. It can be set that the start visualization is shown after logout which might be useful if there is a page which can only be opened by special users.

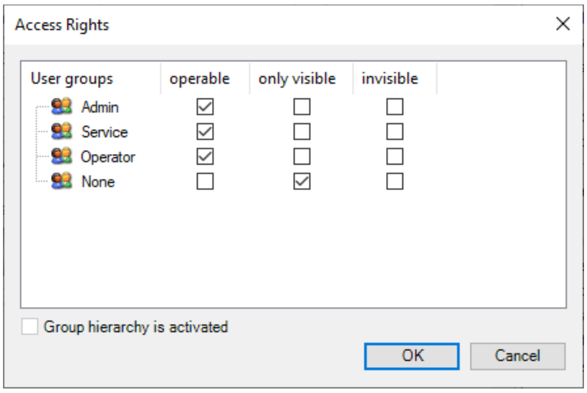
In the Login Settings restrictions can be applied for example after three invalid authentications there is a temporary lockout time of one minute before the user can try to log in again.

16.2 Visualization Configuration

After the user management is enabled each element in the visualization can be configured. In the example page are three buttons.

By selecting the Operator Toggle Button, you can see in the properties that the access rights are limited.

The access rights can be selected and changed. Here the Operator Button is operable for Admin, Service and Operator. None, which means nobody is logged on, can see the button and its state but not change it.

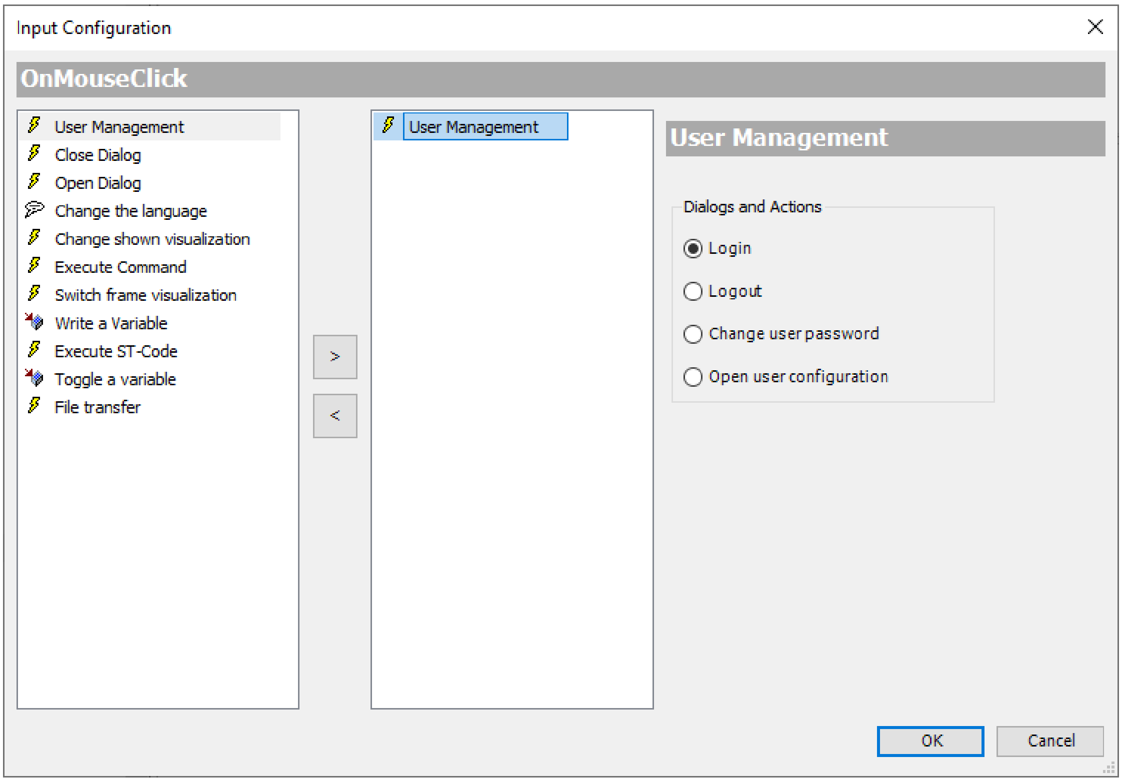


As no Group hierarchy is activated the configuration has to be done for each user group. Otherwise it would be enough to allow the Operator to operate the button. Then automatically the Service and Admin users have also this permission.

The Service button is operable by Admin and Service, Operators can see but not change it and it is invisible for None. The Admin button is only visible and operable for Admins. All other users cannot see it.

Buttons are used to log in and out during runtime as well as for changing a password or the user configuration.

They can be configured with a on mouse click input configuration.



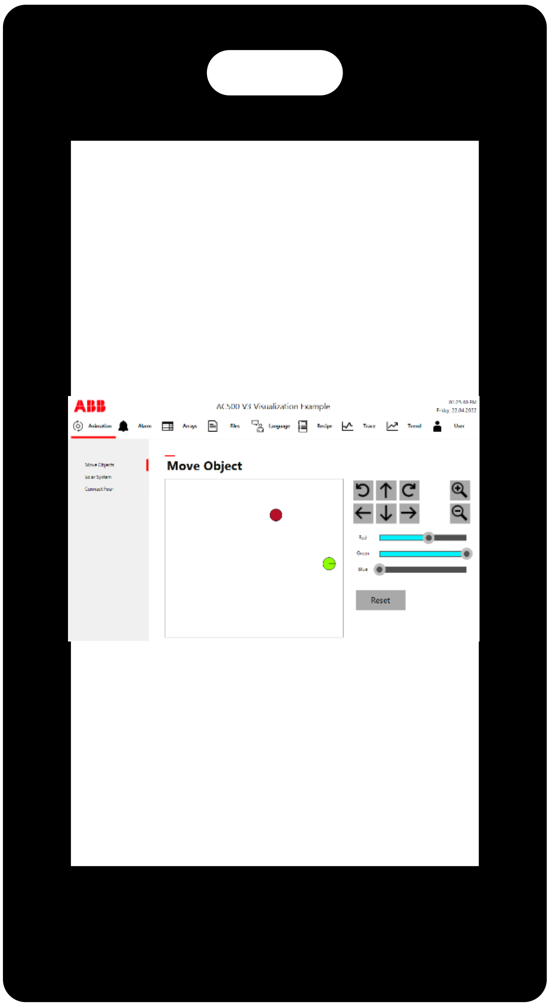
Here Login, Logout, change user password and Open user configuration can be configured.

As in the visualization the usernames and passwords for all users are written the buttons for changing a password or open the user configuration are invisible. To see the buttons in the visualization, remove the TRUE flag in the State variables Invisible.

It is also possible to show inside the visualization which user is logged in and which role he has. The strings are available in the Visu Native Control Library. Use *VisuNativeControl.CurrentUserName* as Text variable to get the logged in user like “Tina”. Use *VisuNativeControl.CurrentFullUserName* as Text variable to get the logged in full user name. Use *VisuNativeControl.CurrentUserGroupName* as Text variable to get the logged in user group name like “Service”.

17 Visualization for mobile devices

When opening a web visualization with a mobile device the visualization will be quite small and it is nearly impossible to click a button.



This chapter shows how to create visualizations which can be used for a mobile device and how to detect if the connected client is portrait or landscape.

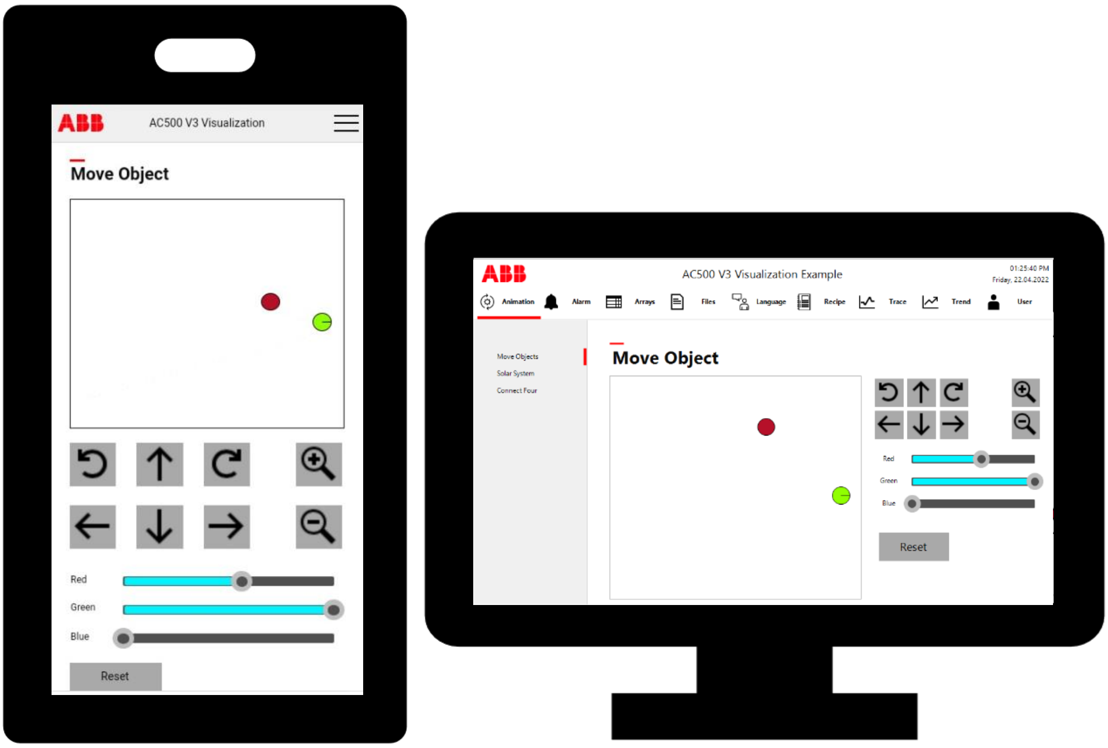
17.1 Creating visualizations in portrait mode

The pages which are already existing for a visualization on a computer browser in landscape mode need to be adapted for the visualization on a smartphone or other devices.

Therefore, all pages are duplicated, and the elements are ordered in a portrait instead of a landscape layout. It might be necessary that some elements are not shown in the portrait view or that some landscape pages are split into multiple portrait pages. Furthermore it might be necessary that the size of some elements is changed.

	Note: When using animation with absolute or relative movement the movement might have to be scaled as the resolution of the field has changed.

When opening the visualization depending on the device either the visualization in landscape or in portrait mode shall be shown.



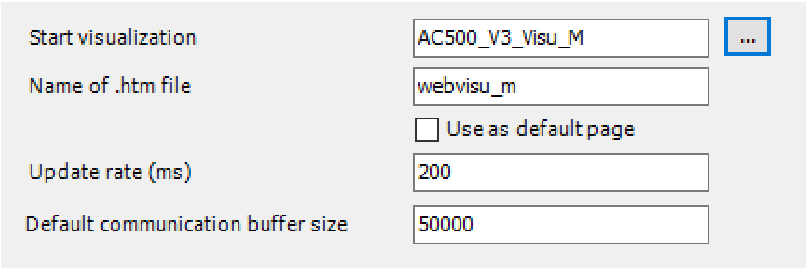
To archive this, two possibilities are shown in this document.

1. Different WebVisu objects below the visualization manager
2. Dynamically screen resolution check with a client manager listener

The first possibility is easy to implement but has the disadvantage that during runtime it cannot be changed between landscape and portrait mode which might be useful when working with a tablet. The second possibility offers much more features, but it is also more effort to implement this functionality and it increases the load of the PLC a bit more.

17.2 Multiple WebVisu objects in the project

As already described in chapter 3.3 it is possible to add more than one WebVisu object below the visualization manager. This way the first WebVisu object can be used as default for all devices which are connecting to the webserver. An additional WebVisu object can be added for mobile devices in portrait mode.



A mobile device can connect to http(s)://<IpAddress>/webvisu_m.htm. Then the AC500_V3_Visu_M page in portrait mode is shown as start visualization instead of AC500_V3_Visu in landscape mode.

17.3 Responsive design

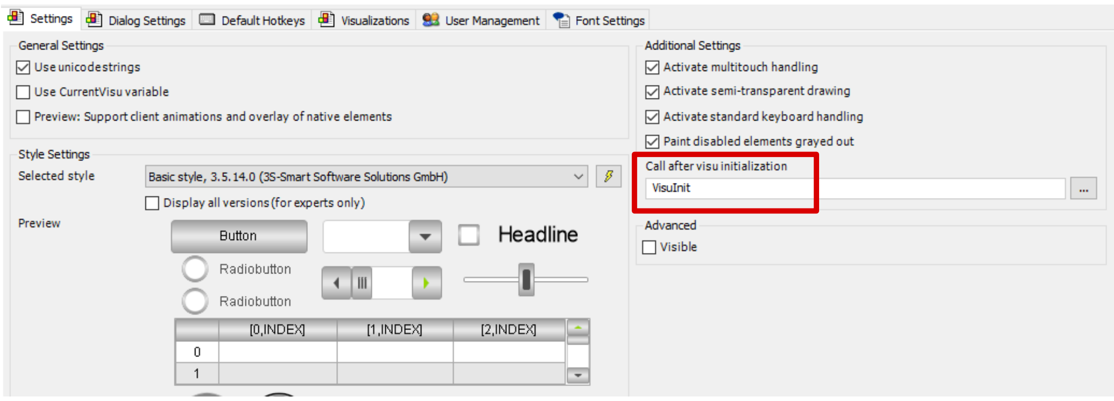
The descriptions in this chapter are based on the example “ResponsiveDesign.project” from the CODESYS store.

The goal of responsive design is to build web pages that detect the visitor’s screen size and orientation and change the layout accordingly.[1] To archive this the resolution of the connected device needs to be read. The necessary steps for reading the resolution are described in the sub chapters.

17.3.1 Client manager listener

To recognize if a new client connects to the webvisu a listener is needed.

The listener itself is initialized with the start of the visualization. Therefore, the program “VisuInit” is used. In the visualization manager this program is called after the initialization of the visualization.



Here for each connected client the “ClientManagerListener” function block is registered. This function block has four methods which are called whenever a client connects, disconnects, changes its size or the starting visualization is set for the first time.

These four methods are edited. Whenever a client is created the method “SetCurrentVisu” is called to change the current displayed visualization. In addition, it is also counted how many clients are connected. The “ClientDestroyed” method can be left empty. Here only the connected clients counter is decreased. For the responsive design the method “ClientSizeChanged” is important. Whenever the size has changed this method is called and another visualization can be shown which fits better to the new screen resolution. A reason for a changed

[1] https://whatis.techtarget.com/definition/responsive-design accessed 15.09.2021

size could for example be the rotation mode in a tablet or splitting the computer screen to show multiple programs in parallel. The fourth method is used to exchange the first shown visualization depending on the screen size.

17.3.2 Set current visualization

The function “SetCurrentVisu” is used to change the current shown visualization for a client. As input the client data is delivered to the function. When this data is existing the function checks if the size of the visualization is set in the client data and reads it. Afterwards it is checked if the screen is in portrait mode. If yes, the mobile visualization is shown, if not the visualization in landscape mode is shown.

> Note: The visualization always changes to the specified visualization. When there is any sub page opened in landscape mode and then the size is changed to portrait mode it will load the home visualization and not the sub page which was opened in landscape mode. When switching back it will continue on the page where the landscape mode had been left.

17.3.3 Client handling

The client handling function block is just reading out some information from the client data. With the method “GetClientSize” it is reading the coordinates from the bottom right corner. As the upper left corner has the coordinates 0;0 these gives the resolution of the screen. With the method IsPortrait it is checked whether the height is greater than the width. If this is the case the method returns true which means the portrait visualizations will be shown later.

It could also be possible to make a more detailed analysis of the screen resolution and change between more than two different visualization formats.

ABB AG
Eppelheimer Straße 82
69123 Heidelberg, Germany
Phone: +49 62 21 701 1444
Fax: +49 62 21 701 1382
E-Mail: plc.support@de.abb.com
www.abb.com/plc